

BFS Traversal

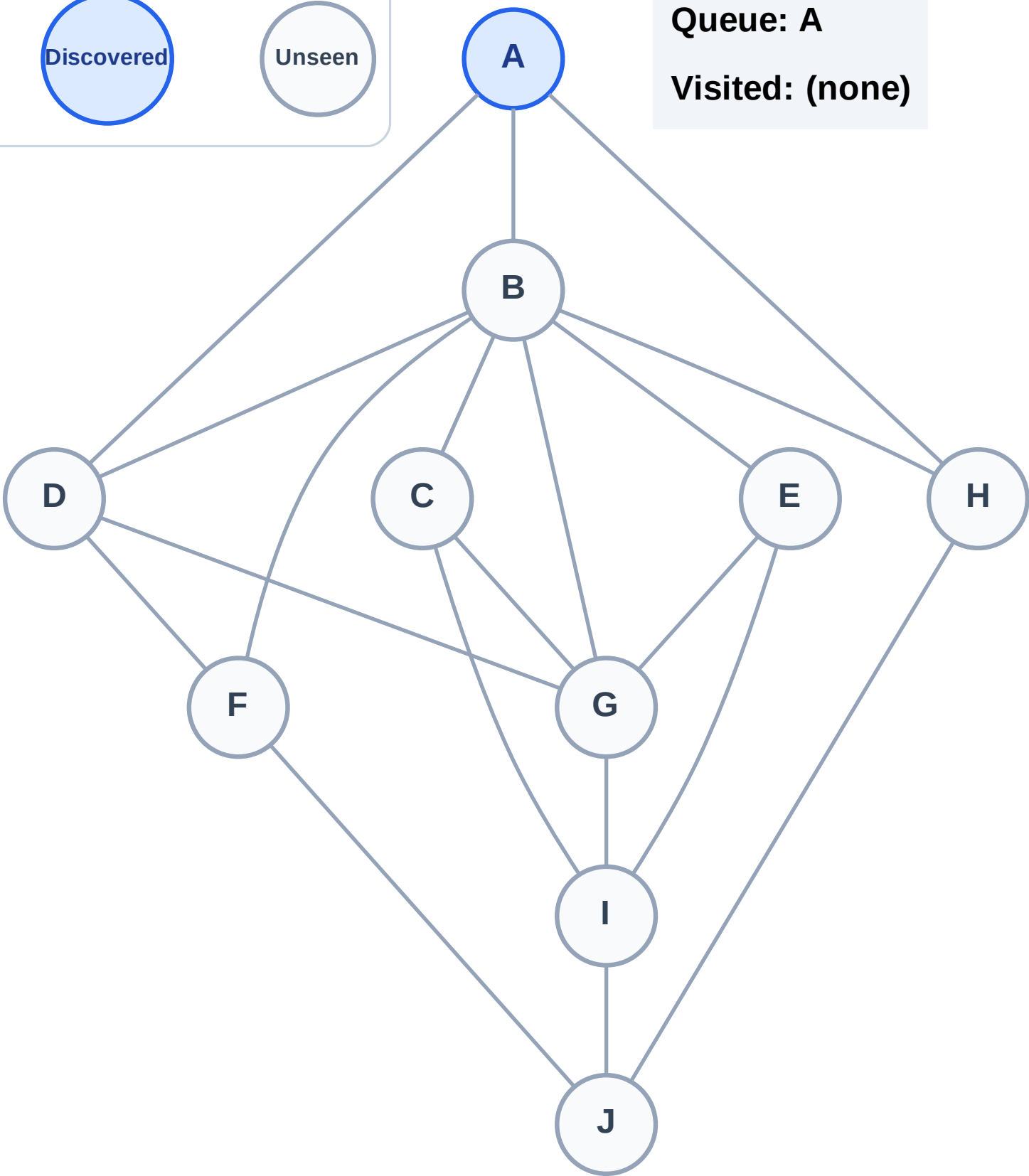
Step 1: Start at A

Legend



Queue: A

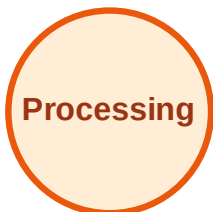
Visited: (none)



BFS Traversal

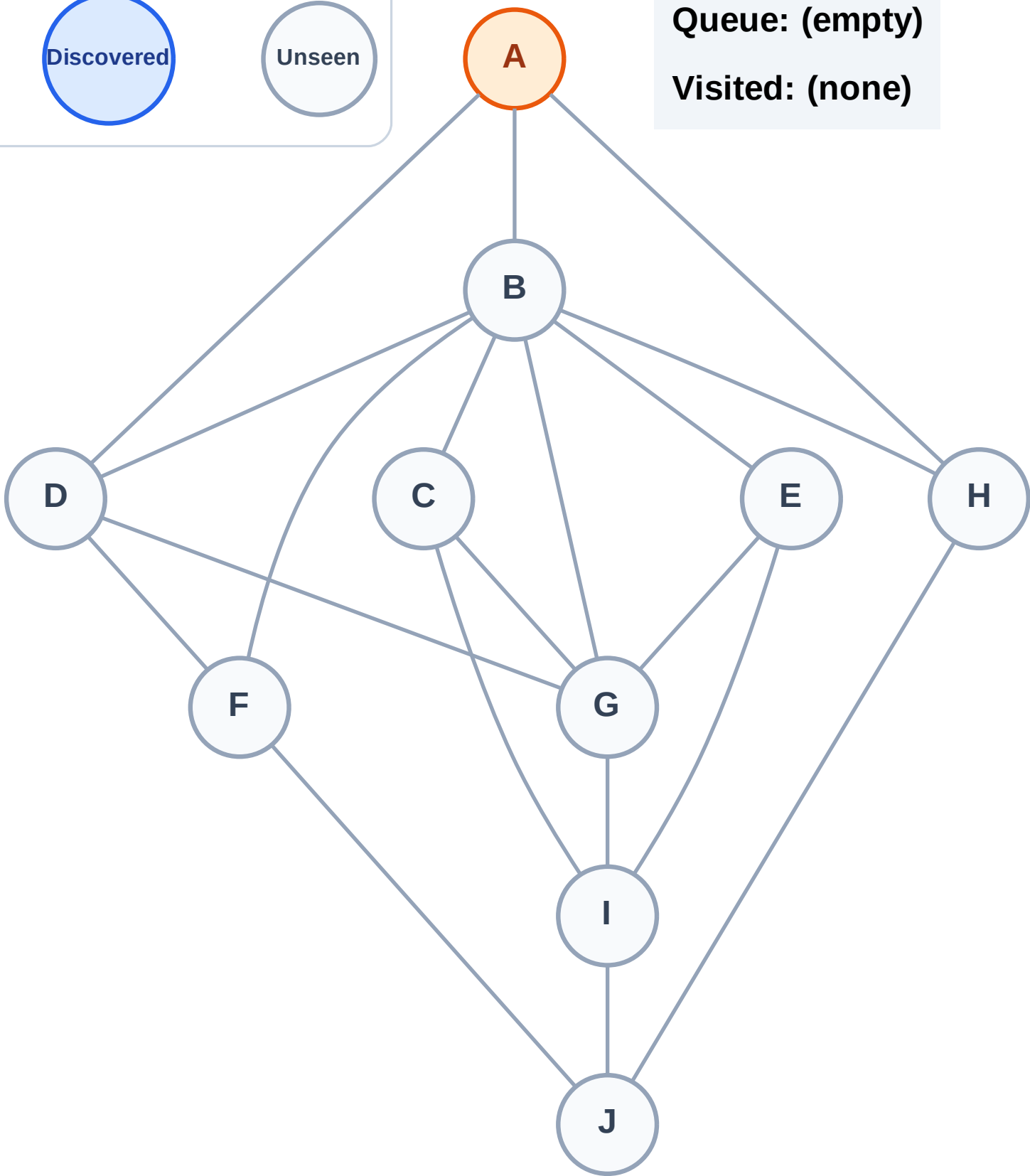
Step 2: Process node A

Legend



Queue: (empty)

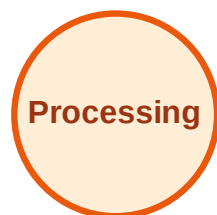
Visited: (none)



BFS Traversal

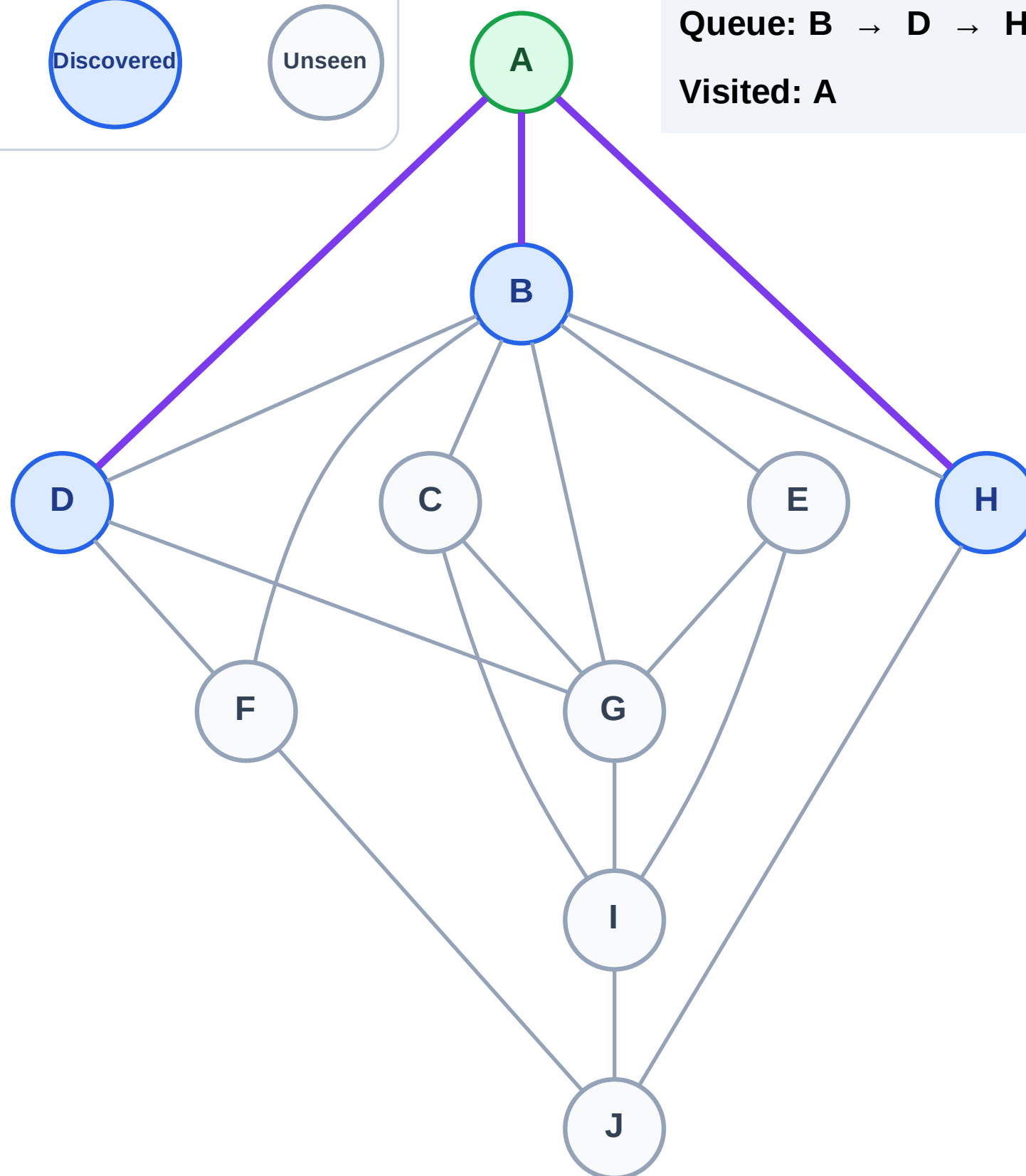
Step 3: Discover B, D, H from A

Legend



Queue: B → D → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

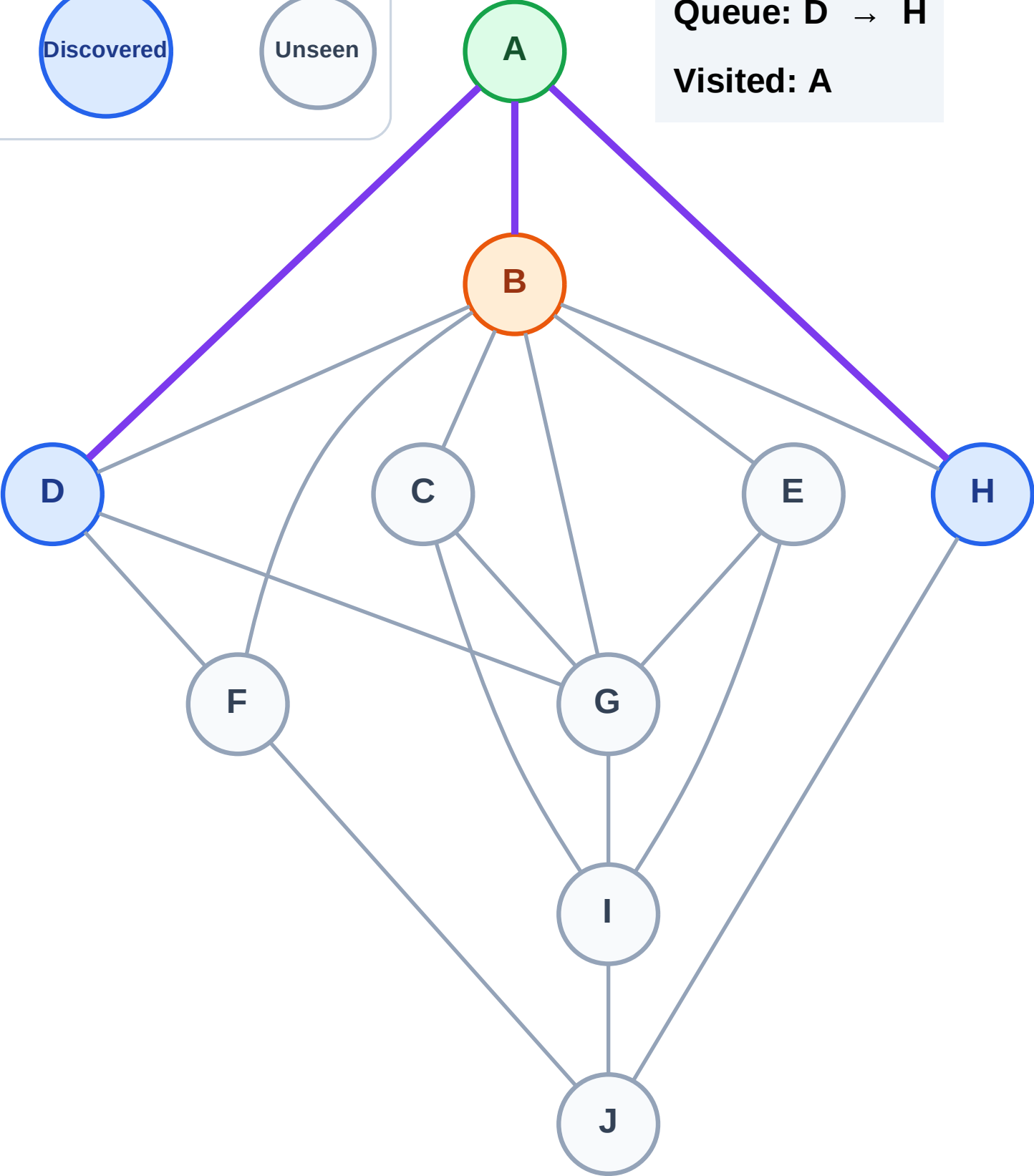
Complete

Processing

Discovered

Unseen

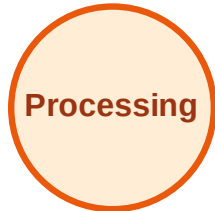
Queue: D → H
Visited: A



BFS Traversal

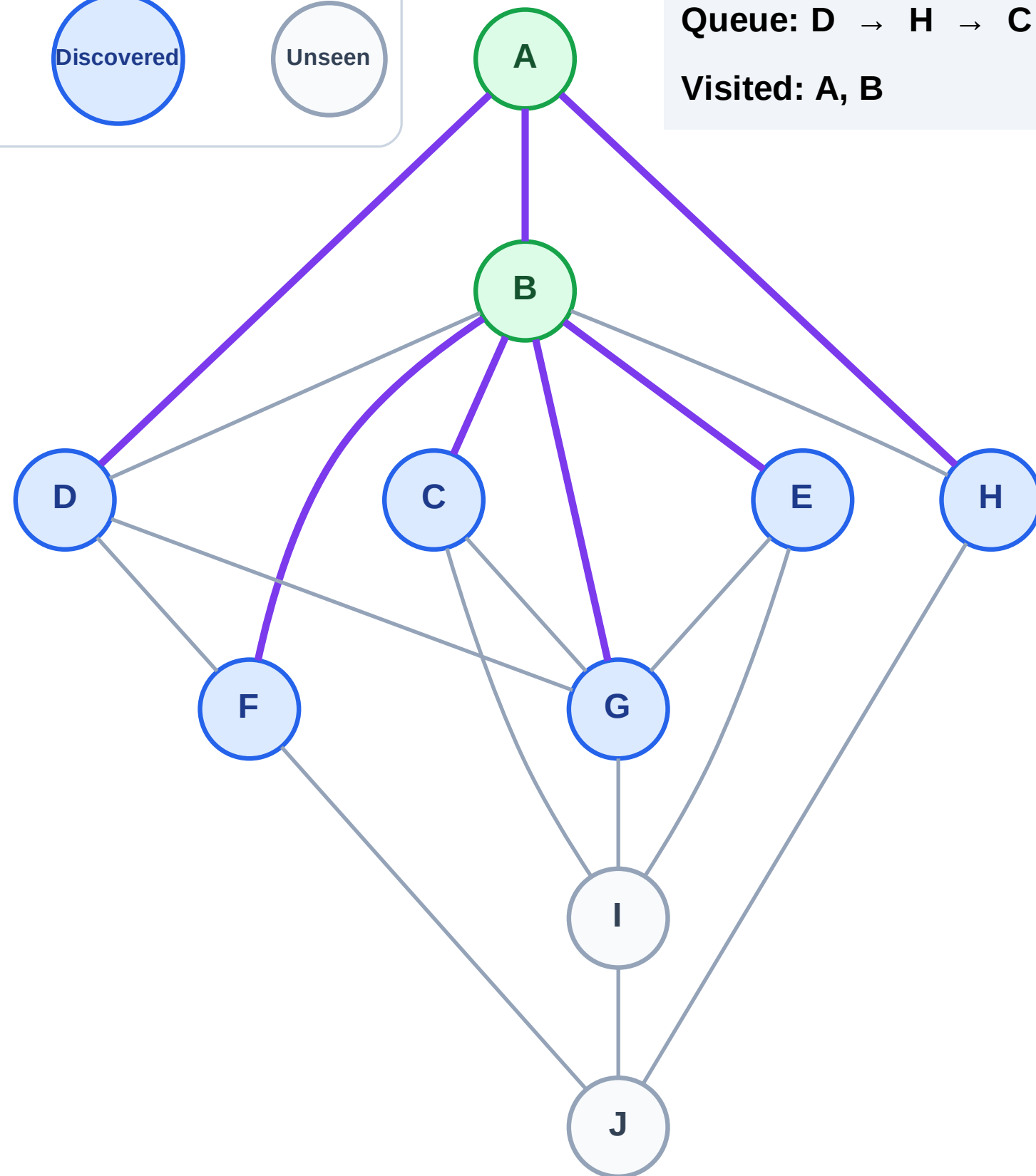
Step 5: Discover C, E, F, G from B

Legend



Queue: D → H → C → E → F → G

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

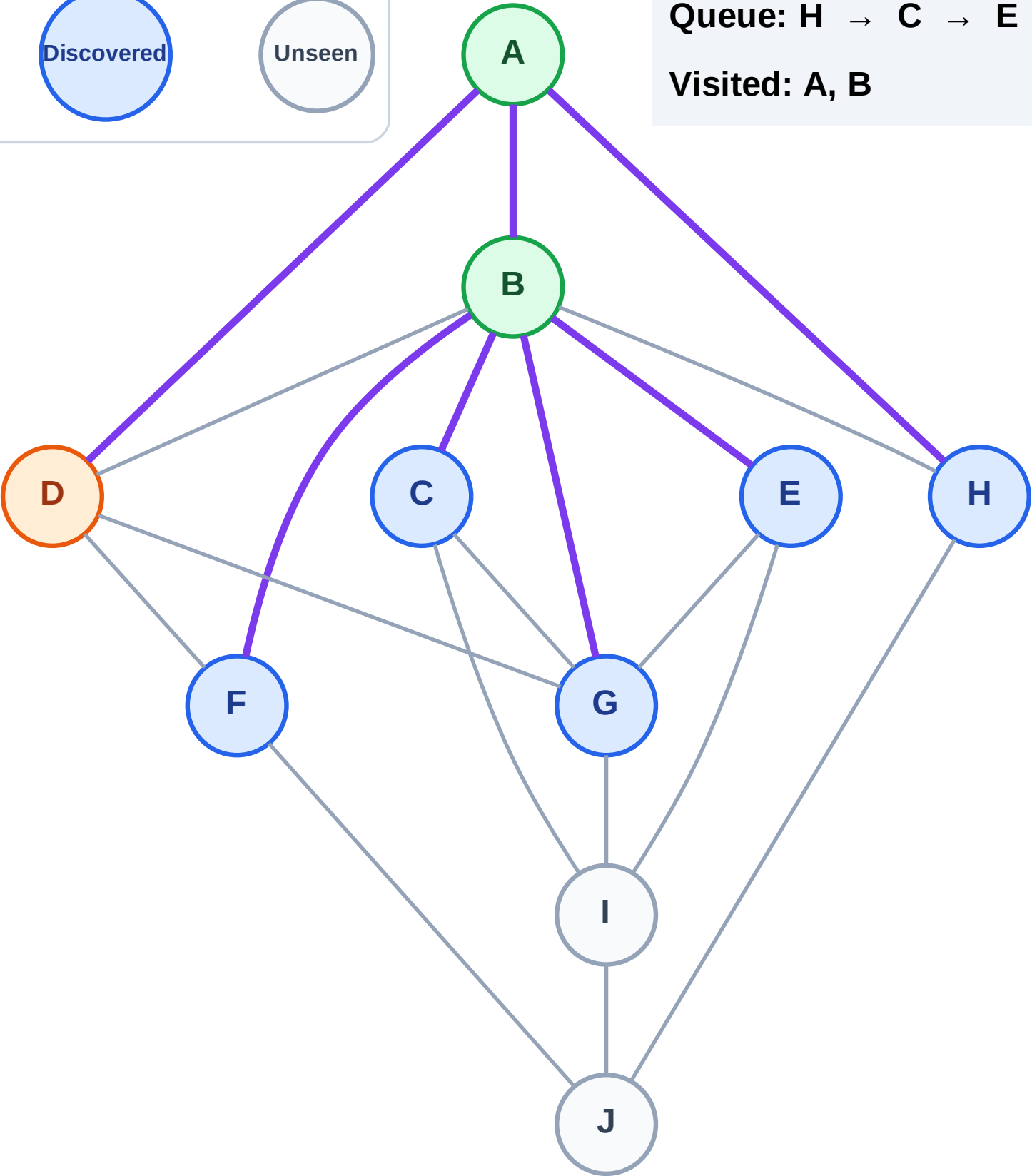
Processing

Discovered

Unseen

Queue: H → C → E → F → G

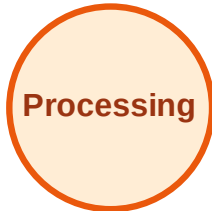
Visited: A, B



BFS Traversal

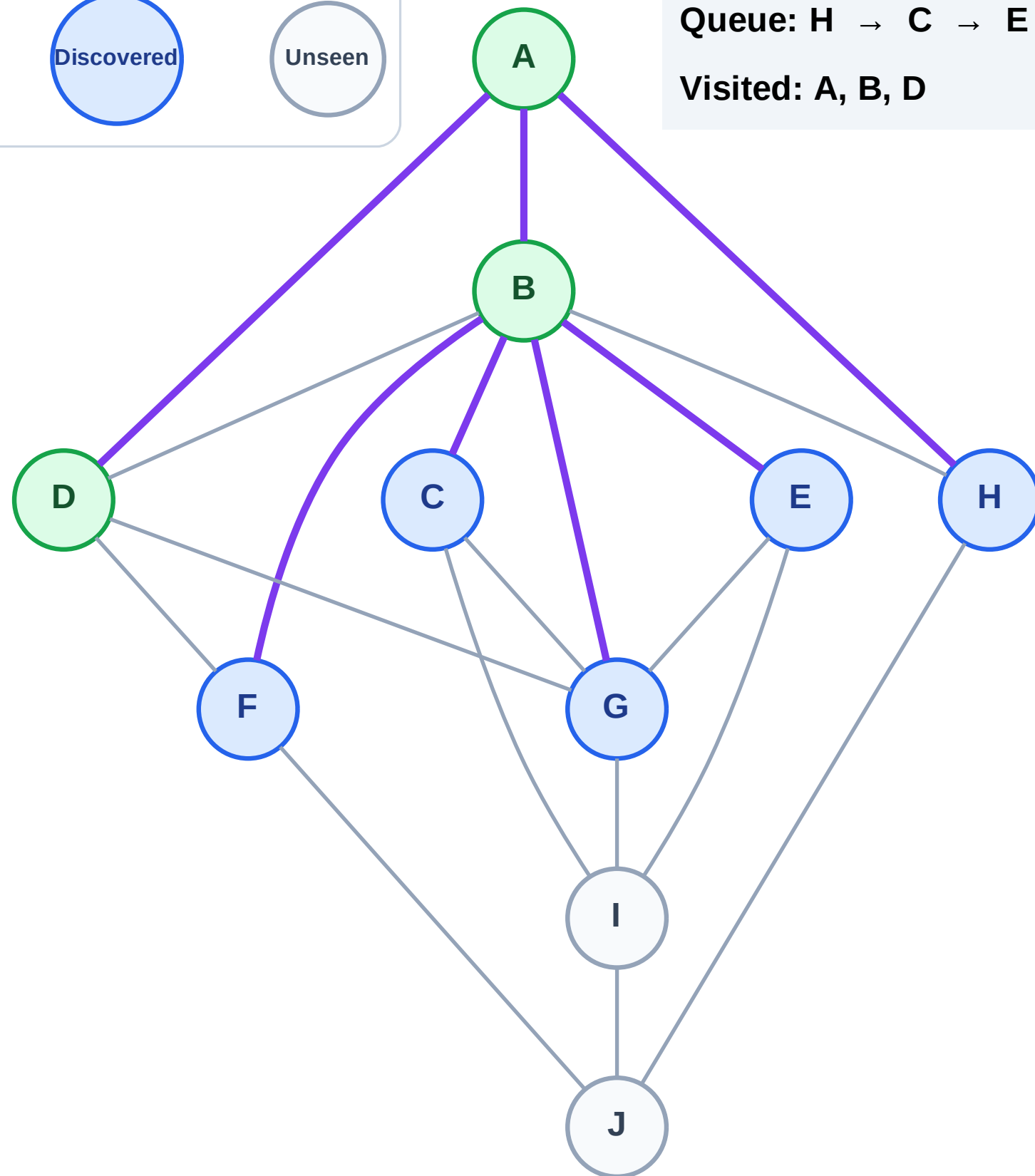
Step 7: No new neighbors from D

Legend



Queue: H → C → E → F → G

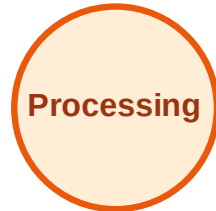
Visited: A, B, D



BFS Traversal

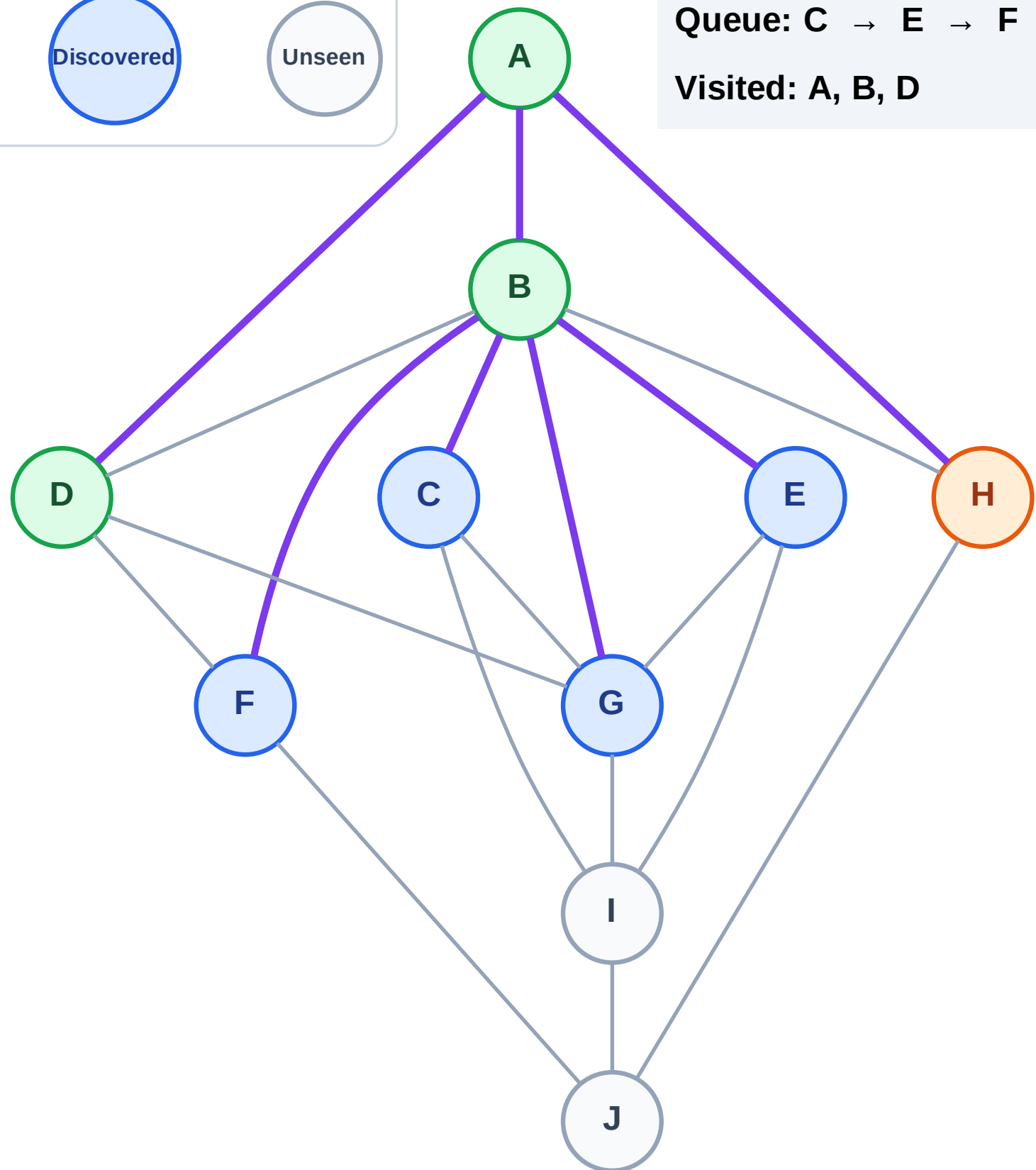
Step 8: Process node H

Legend



Queue: C → E → F → G

Visited: A, B, D



BFS Traversal

Step 9: Discover J from H

Legend

Complete

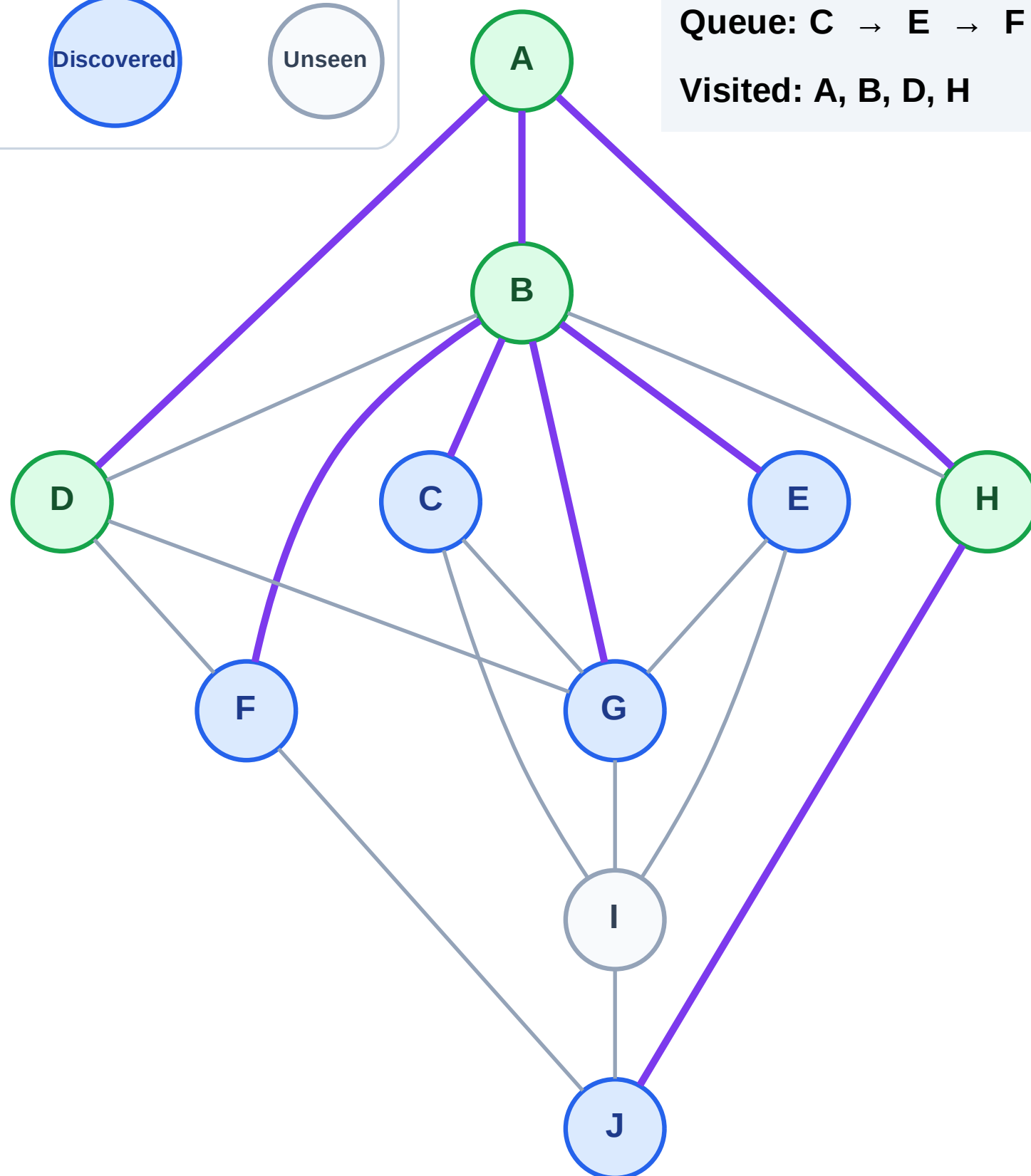
Processing

Discovered

Unseen

Queue: C → E → F → G → J

Visited: A, B, D, H



BFS Traversal

Step 10: Process node C

Legend

Complete

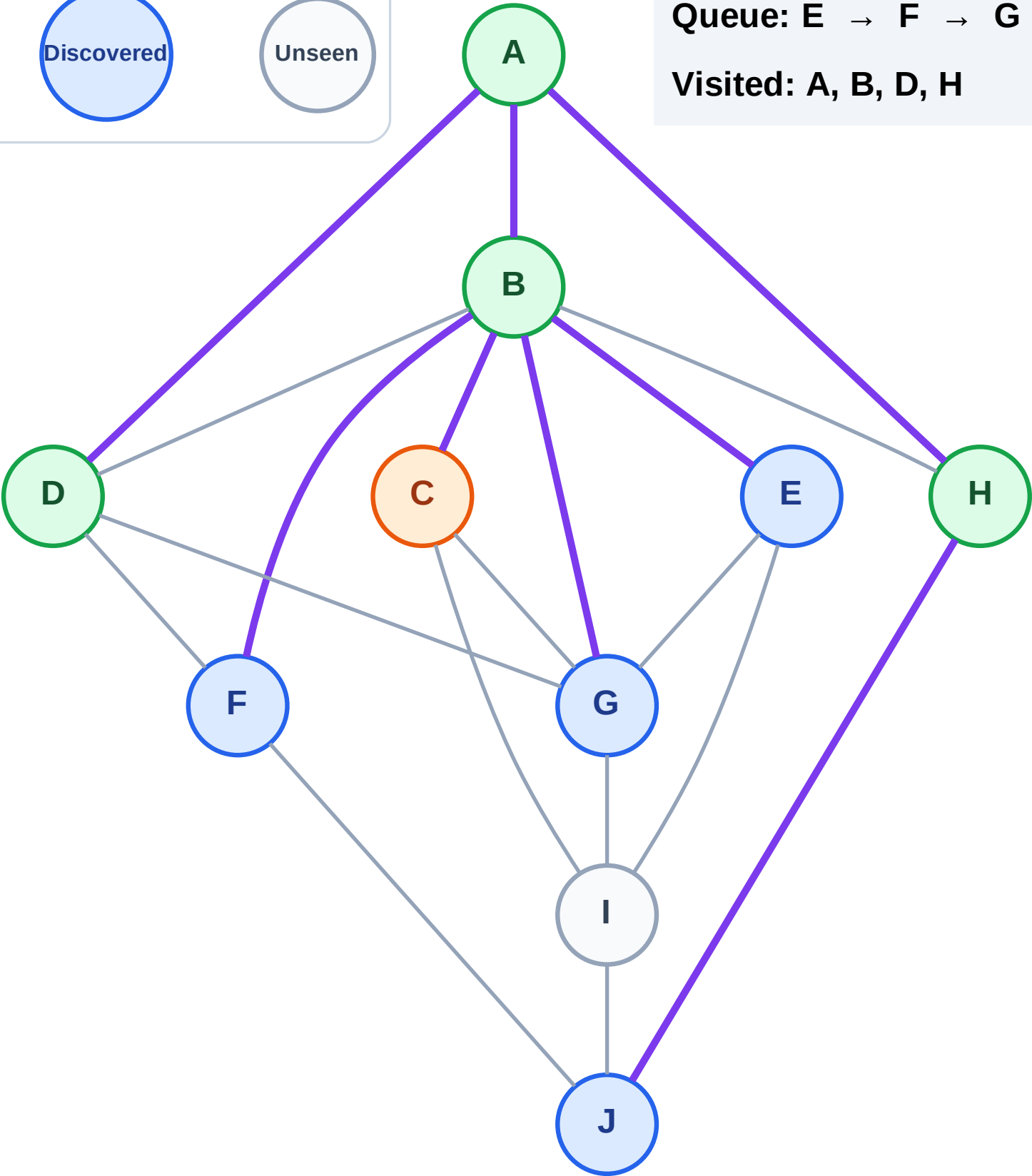
Processing

Discovered

Unseen

Queue: E → F → G → J

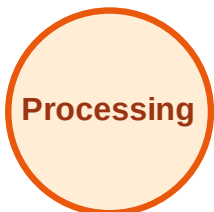
Visited: A, B, D, H



BFS Traversal

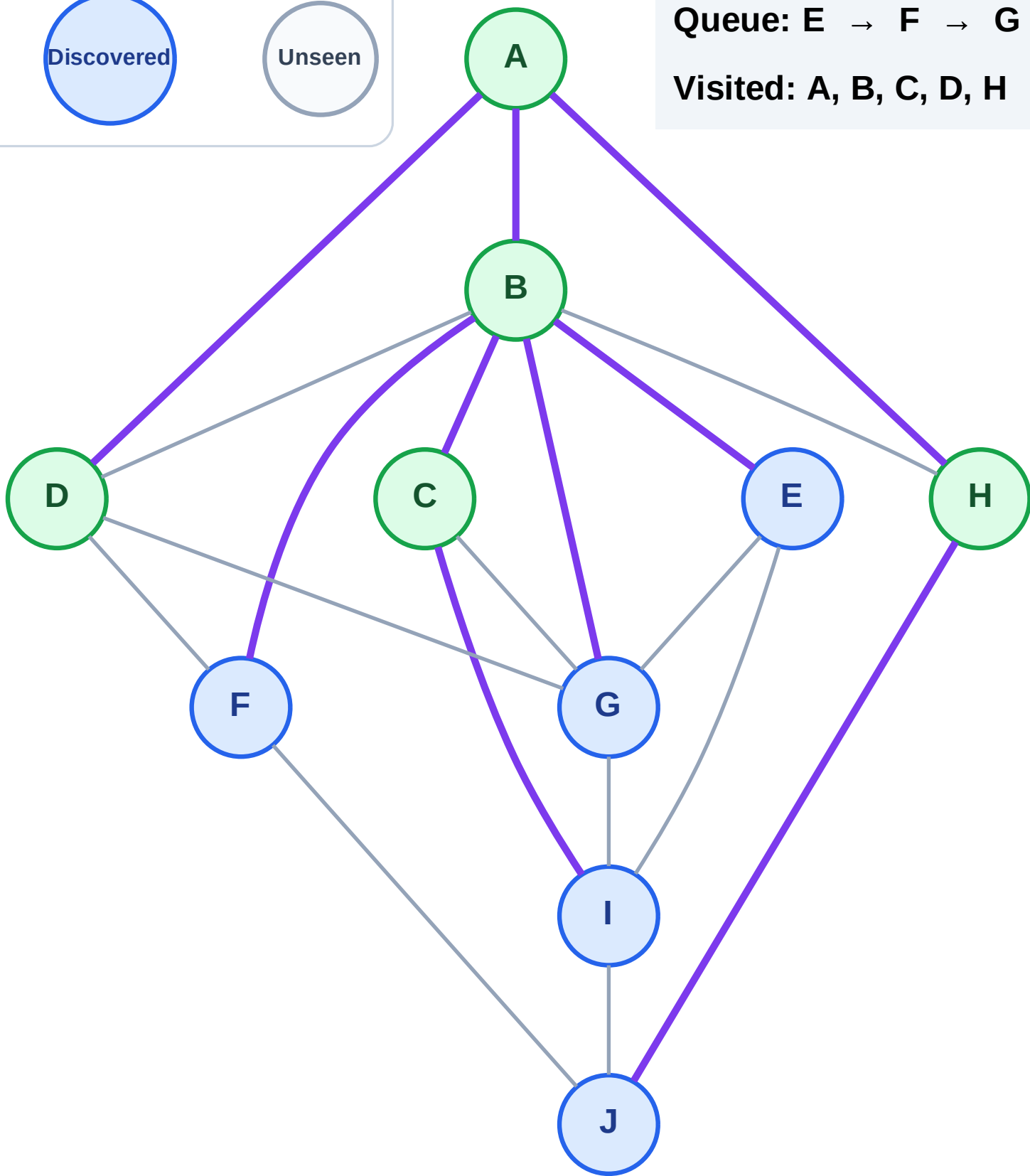
Step 11: Discover I from C

Legend



Queue: E → F → G → J → I

Visited: A, B, C, D, H



BFS Traversal

Step 12: Process node E

Legend

Complete

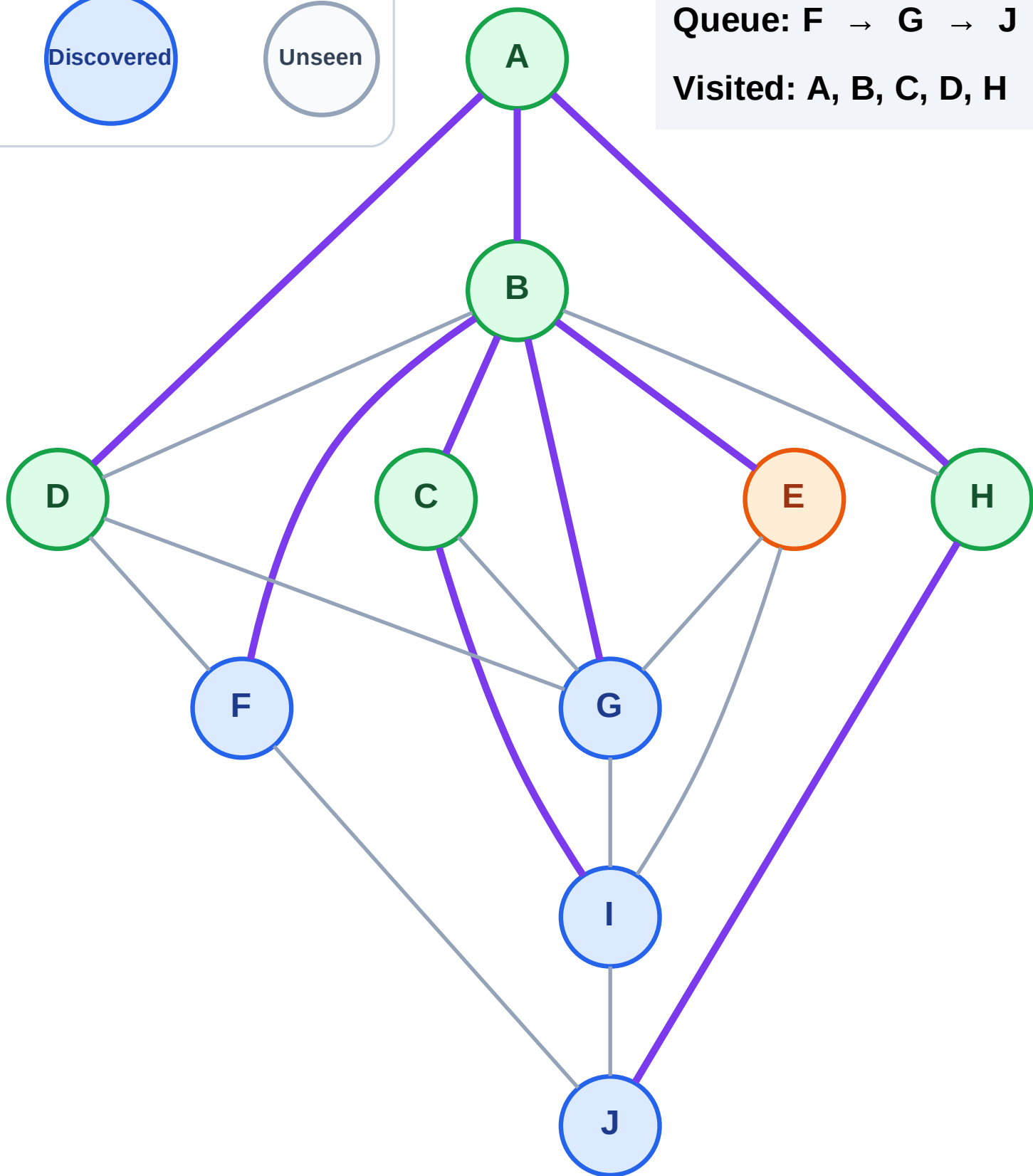
Processing

Discovered

Unseen

Queue: F → G → J → I

Visited: A, B, C, D, H



BFS Traversal

Step 13: No new neighbors from E

Legend

Complete

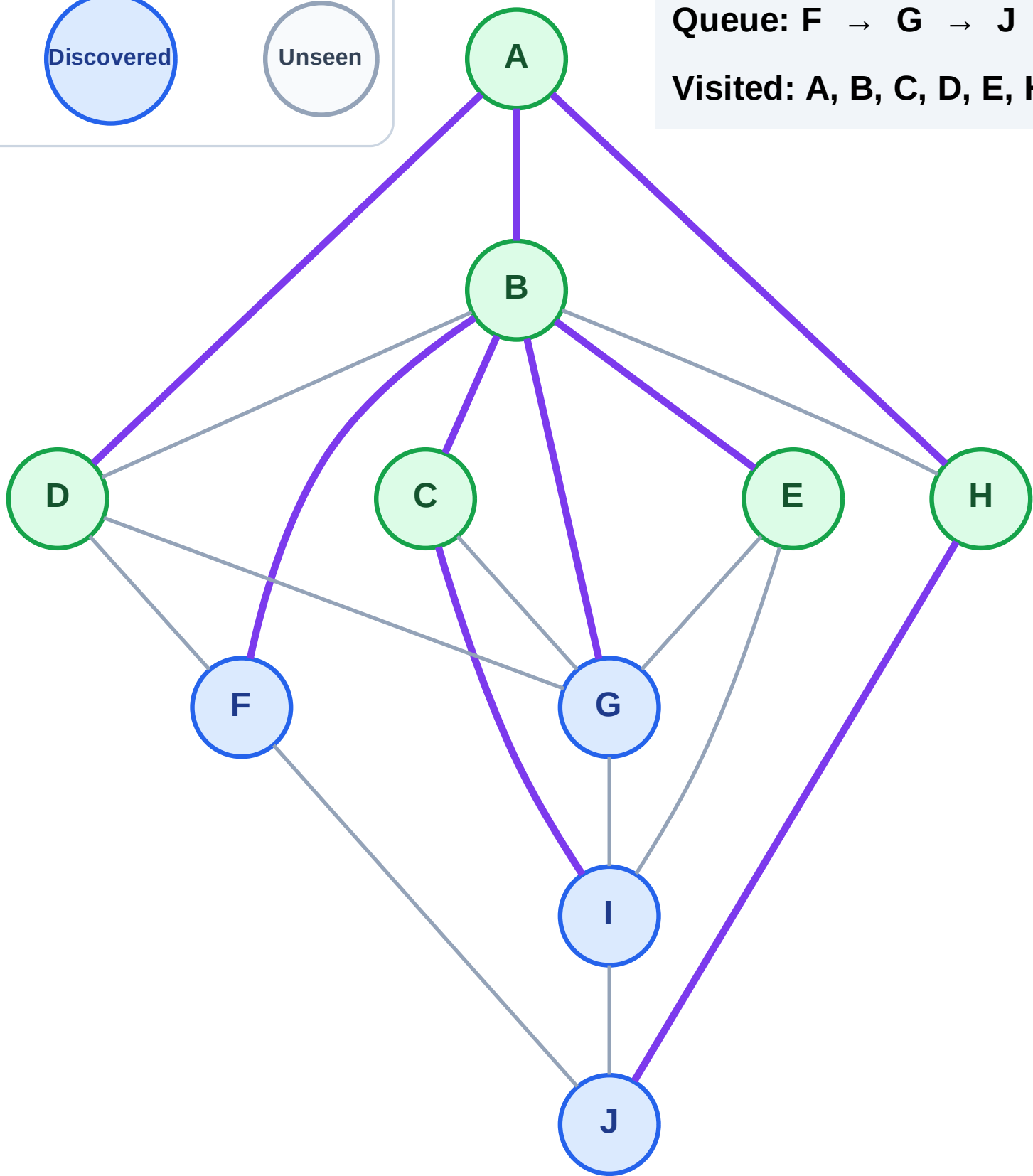
Processing

Discovered

Unseen

Queue: F → G → J → I

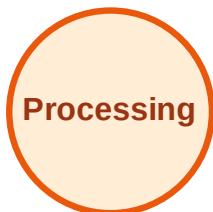
Visited: A, B, C, D, E, H



BFS Traversal

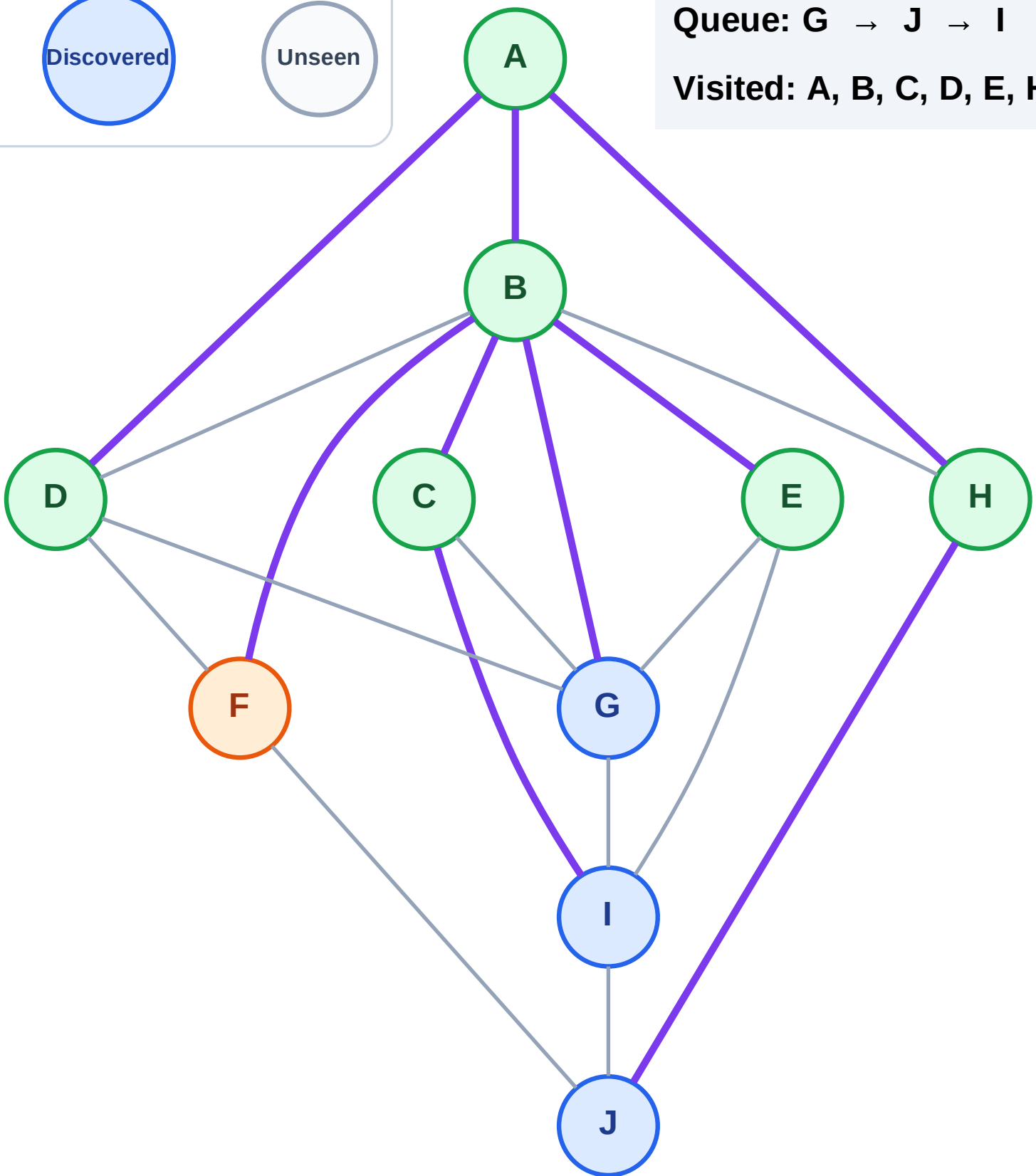
Step 14: Process node F

Legend



Queue: G → J → I

Visited: A, B, C, D, E, H



BFS Traversal

Step 15: No new neighbors from F

Legend

Complete

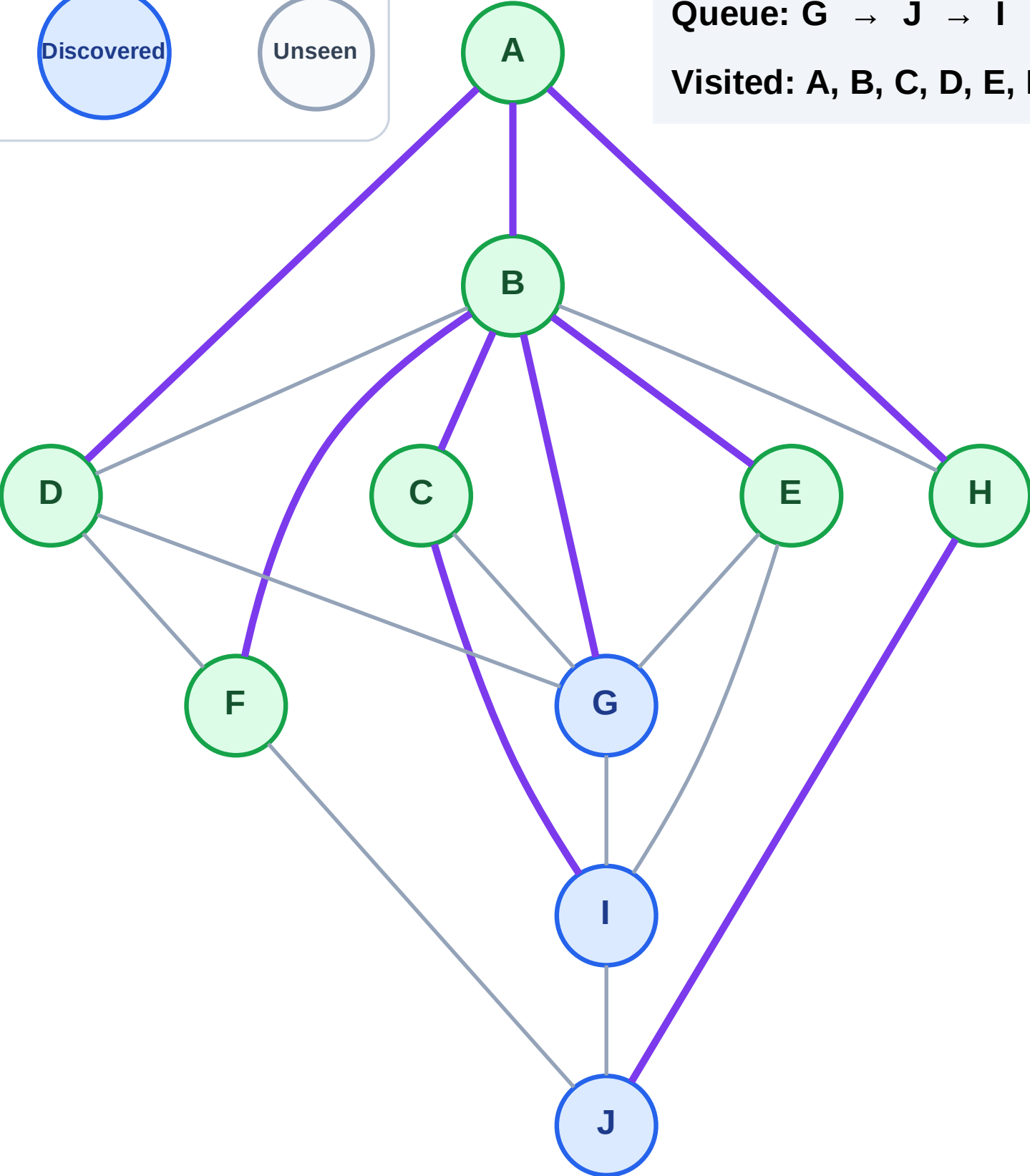
Processing

Discovered

Unseen

Queue: G → J → I

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node G

Legend

Complete

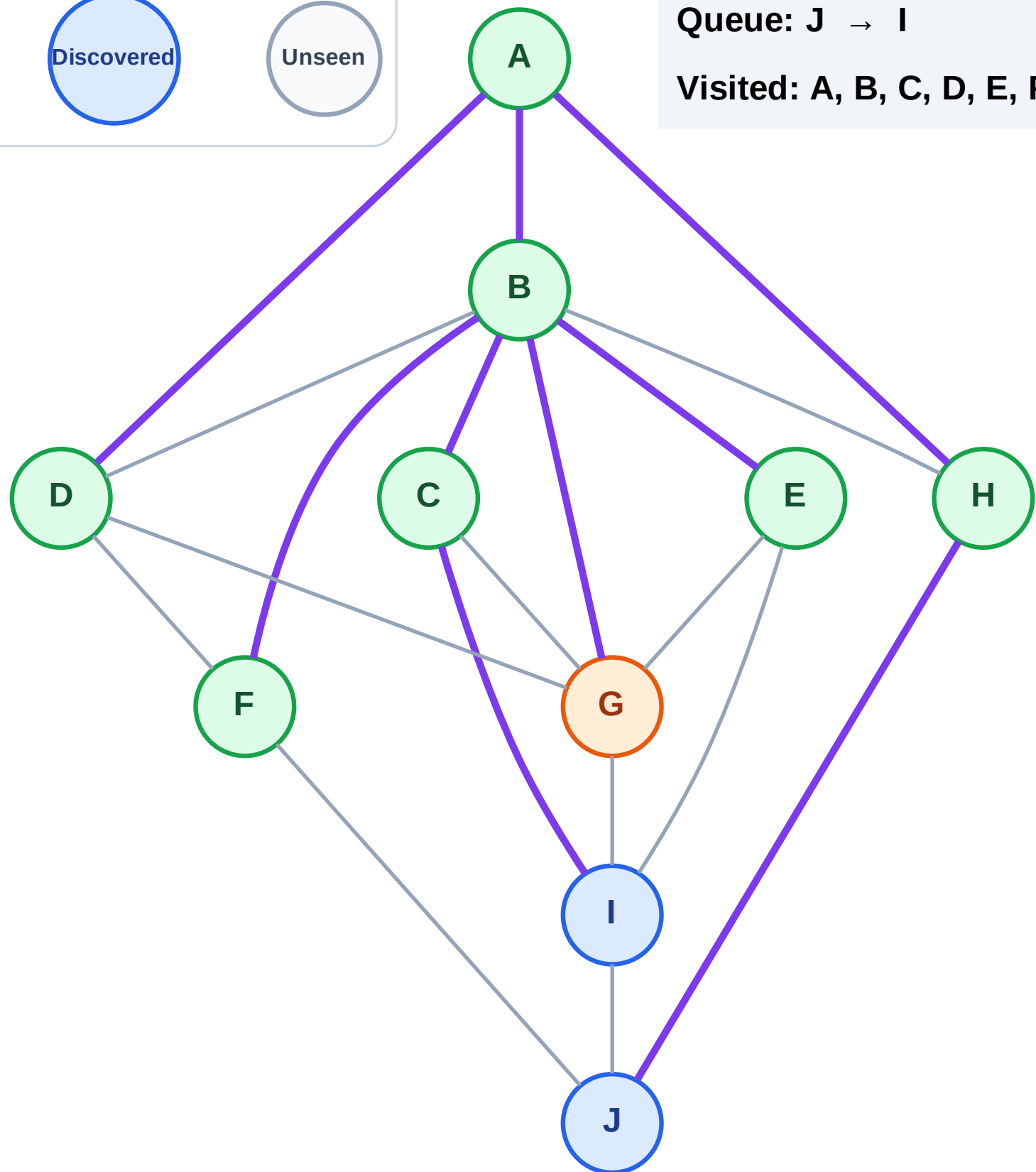
Processing

Discovered

Unseen

Queue: J → I

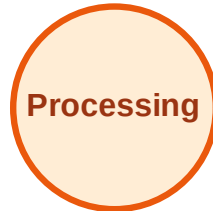
Visited: A, B, C, D, E, F, H



BFS Traversal

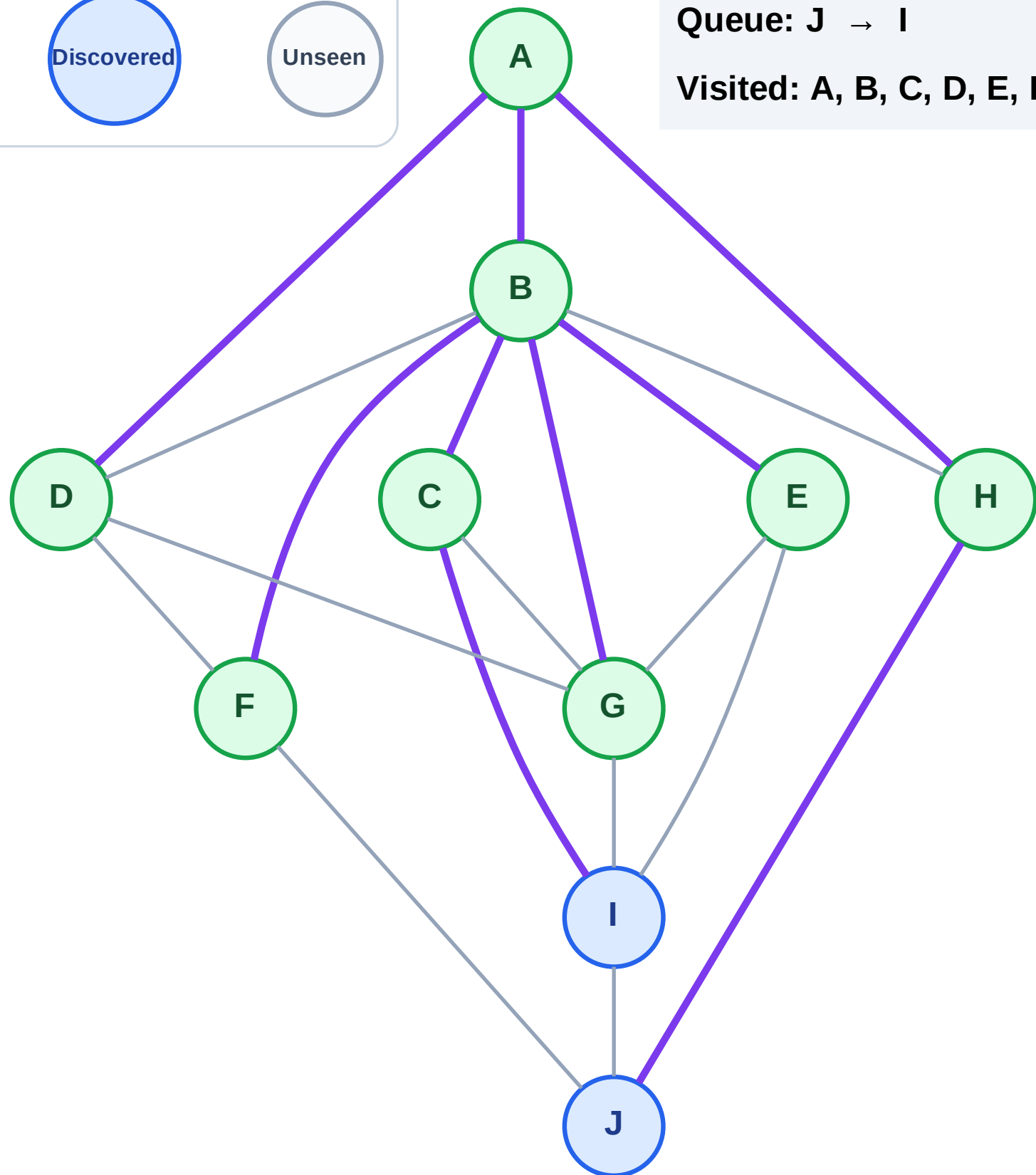
Step 17: No new neighbors from G

Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node J

Legend

Complete

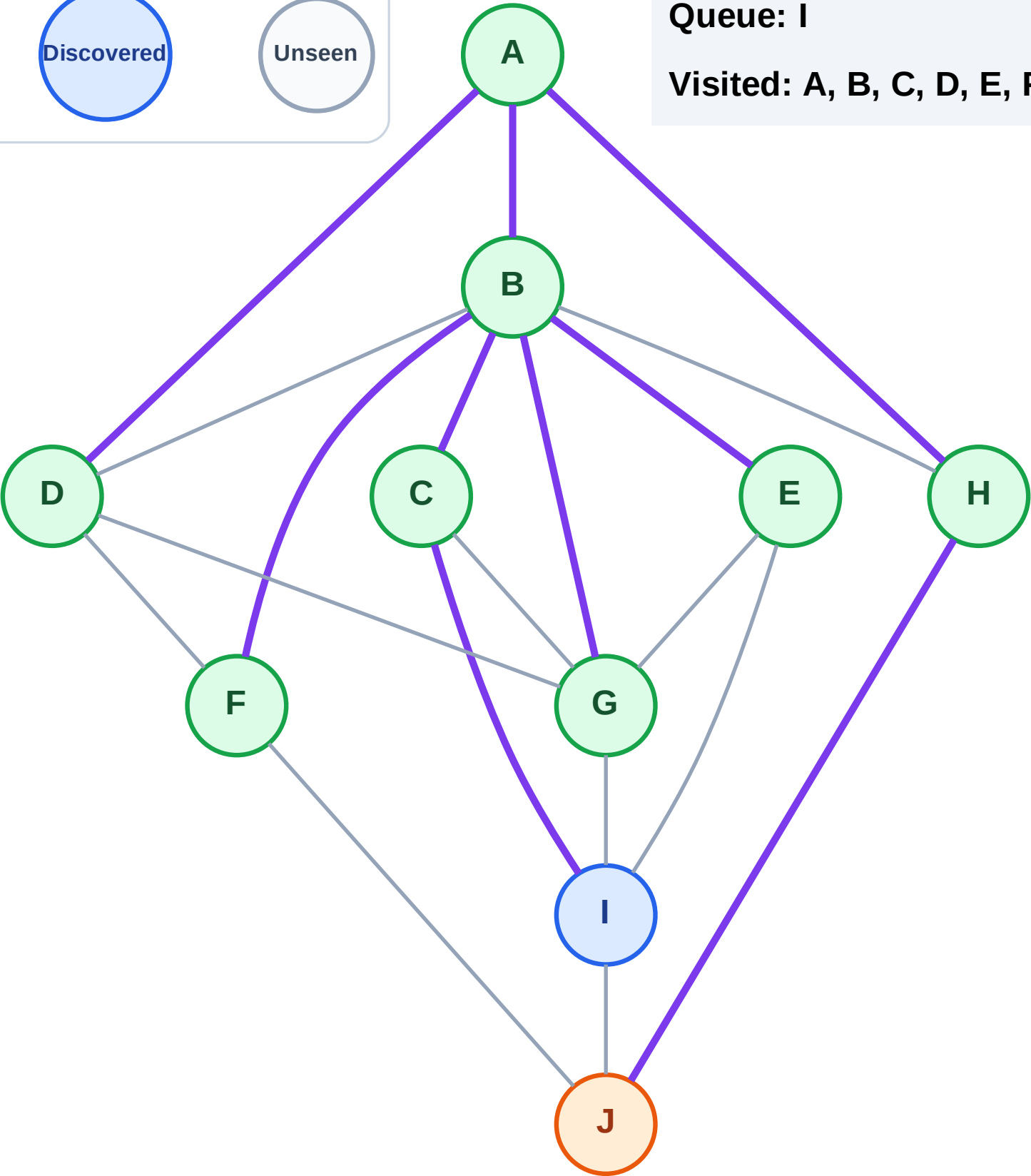
Processing

Discovered

Unseen

Queue: I

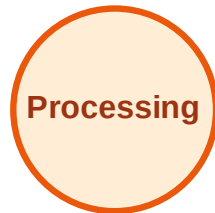
Visited: A, B, C, D, E, F, G, H



BFS Traversal

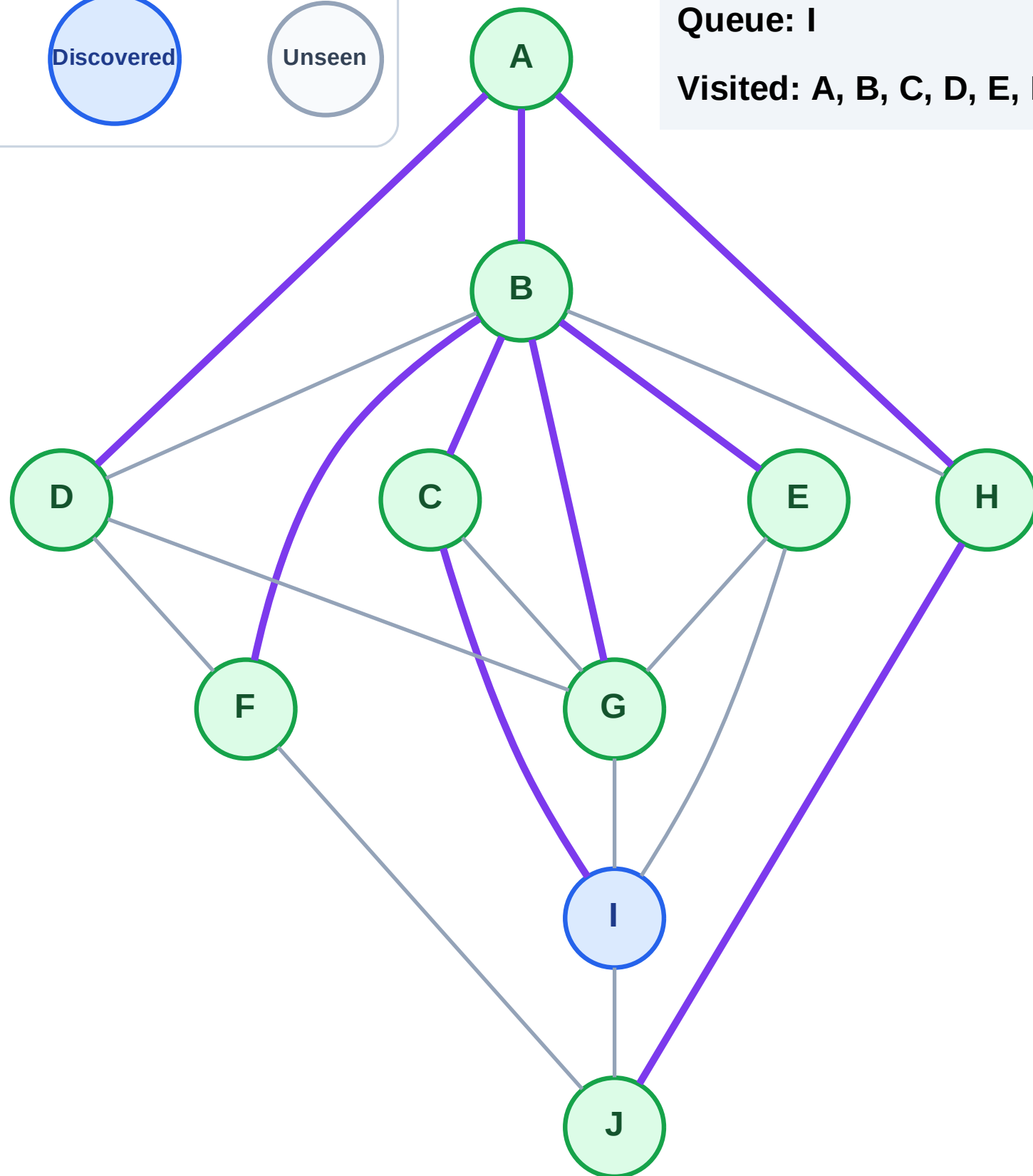
Step 19: No new neighbors from J

Legend



Queue: I

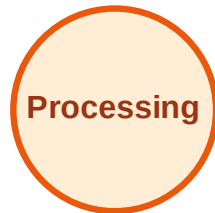
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

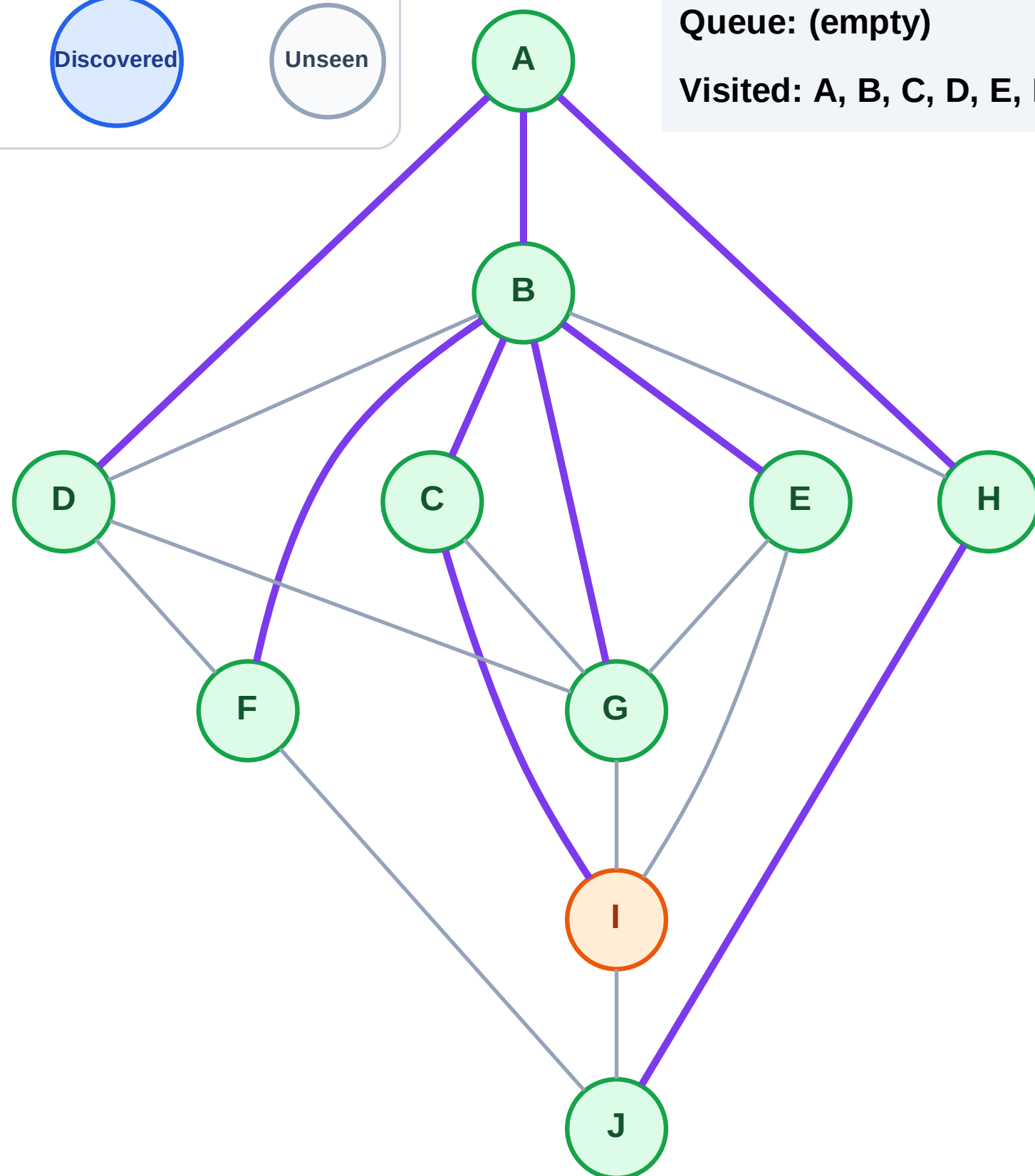
Step 20: Process node I

Legend



Queue: (empty)

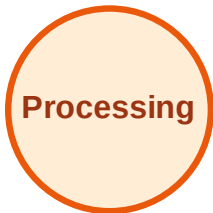
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

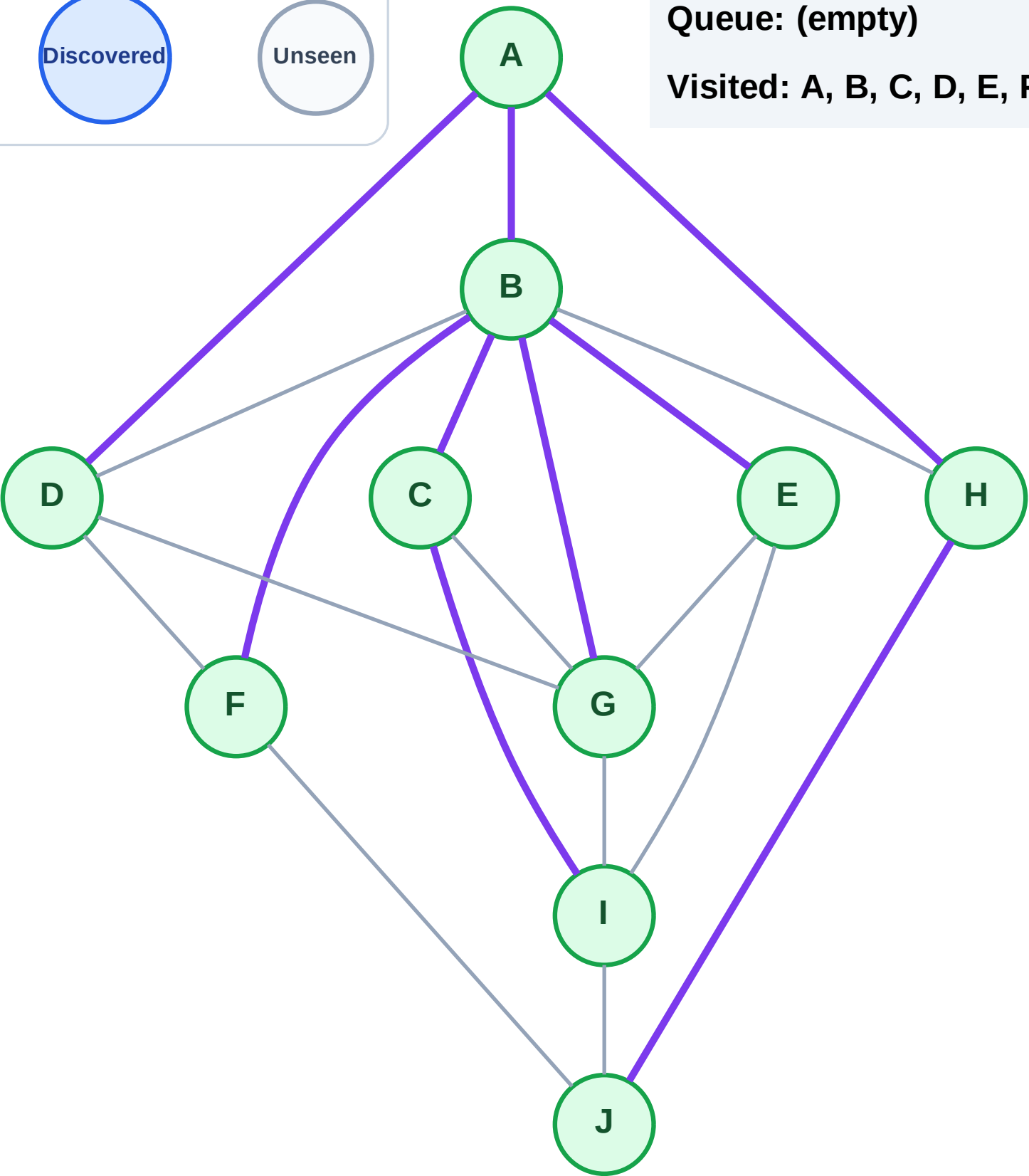
Step 21: No new neighbors from I

Legend



Queue: (empty)

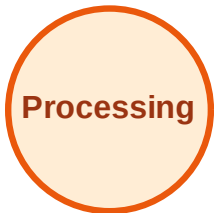
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

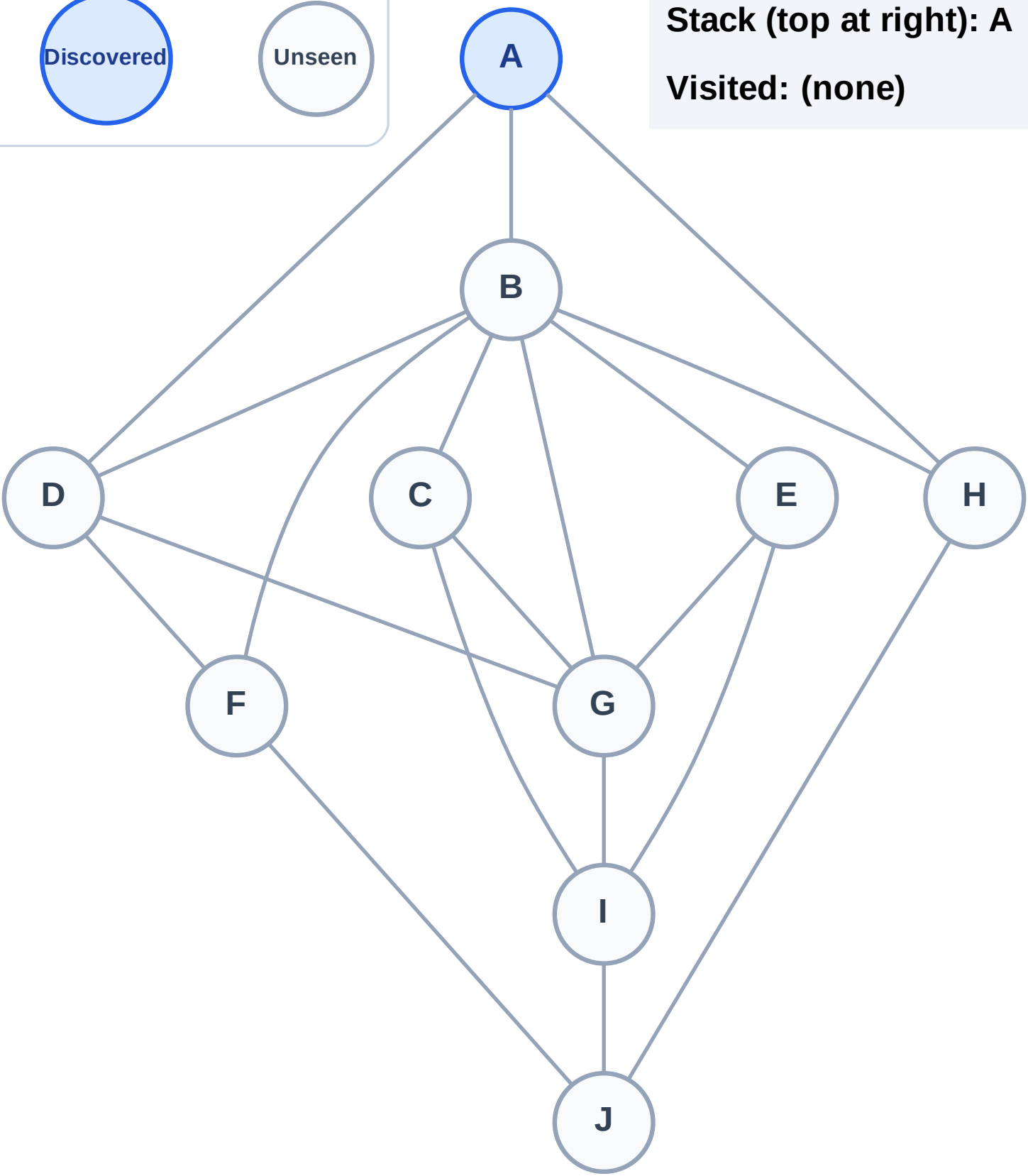
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

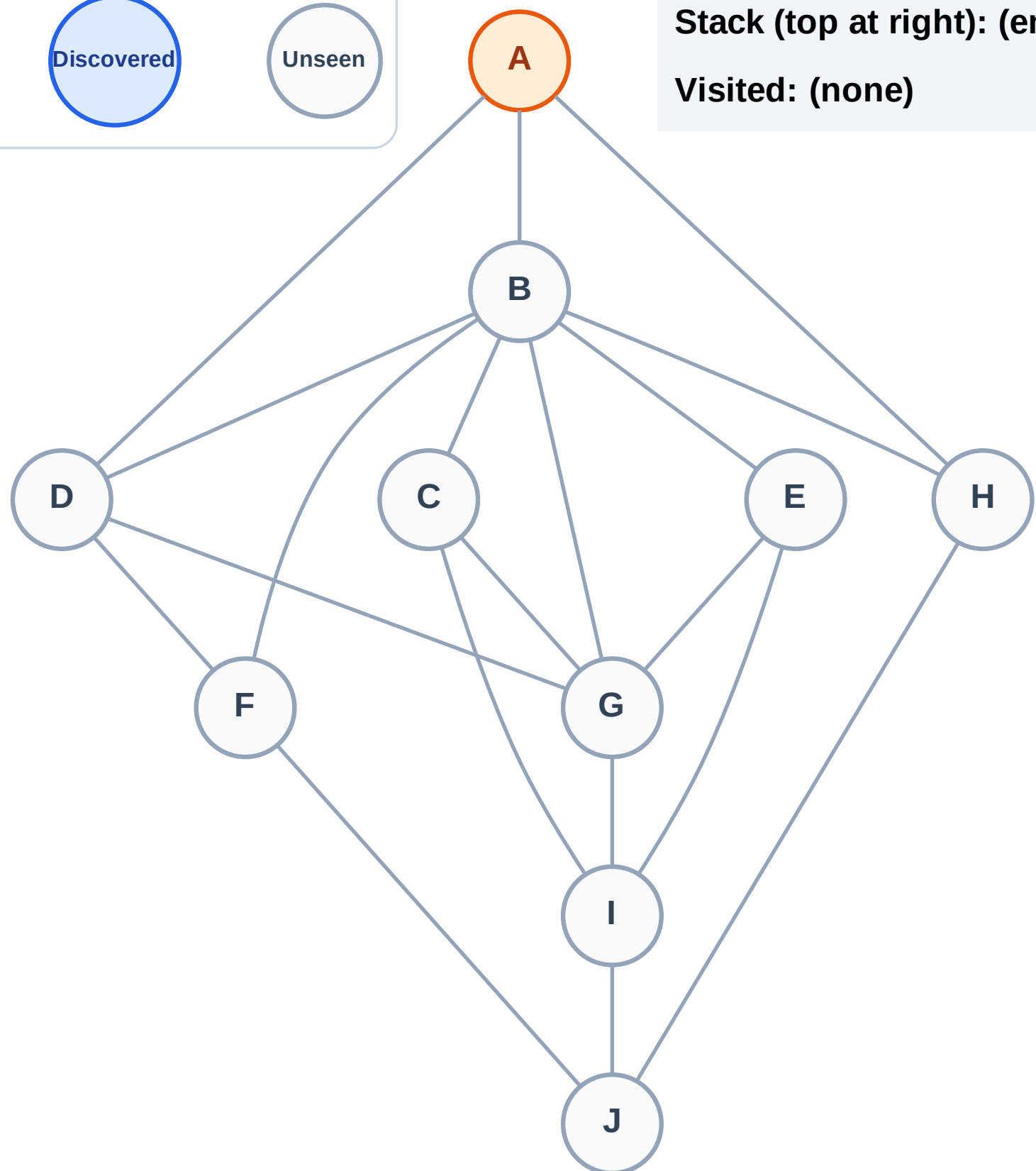
Processing

Discovered

Unseen

Stack (top at right): (empty)

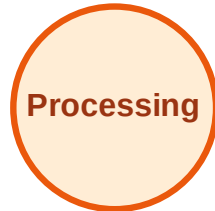
Visited: (none)



DFS Traversal

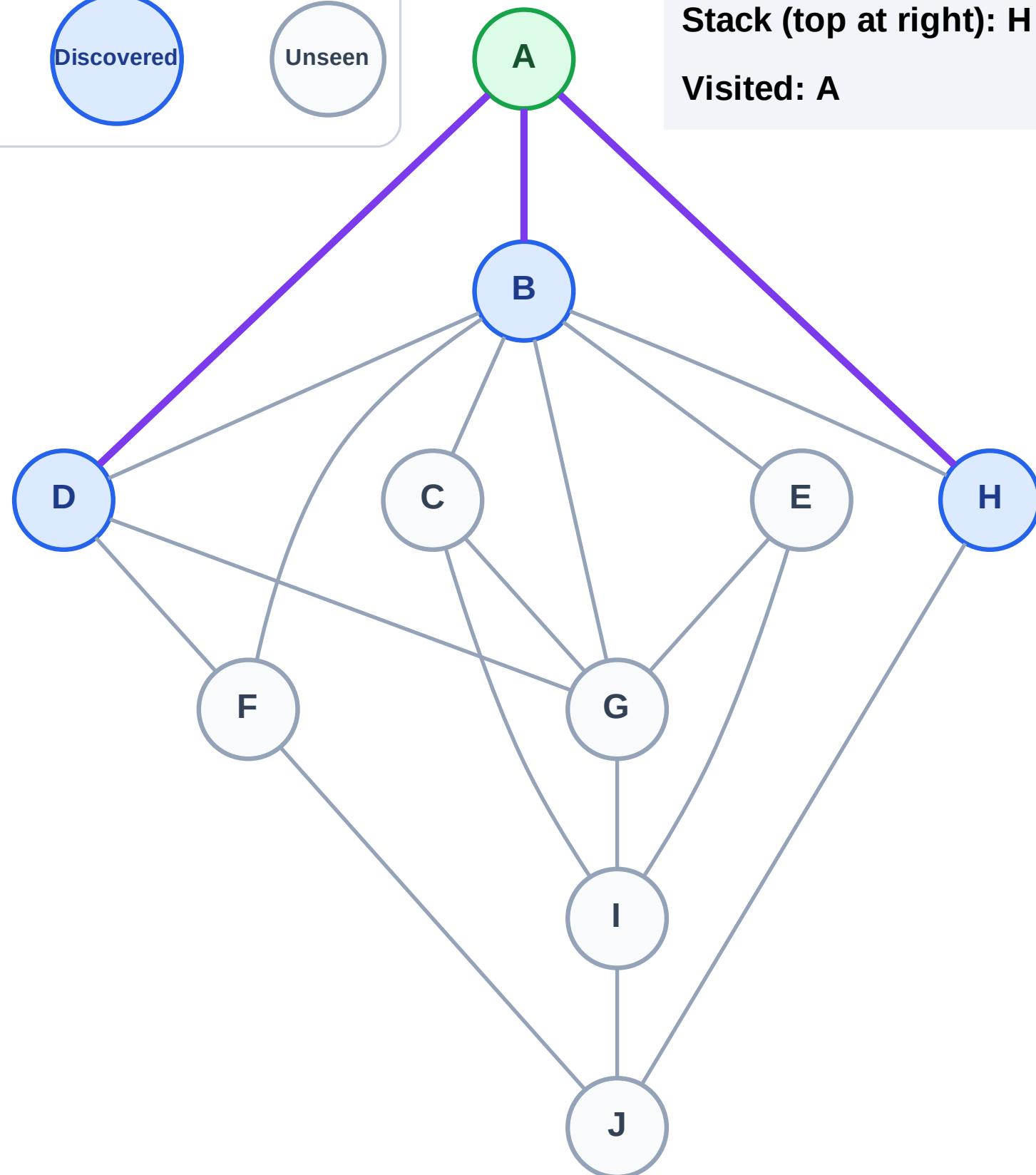
Step 3: Discover H, D, B from A

Legend



Stack (top at right): H | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

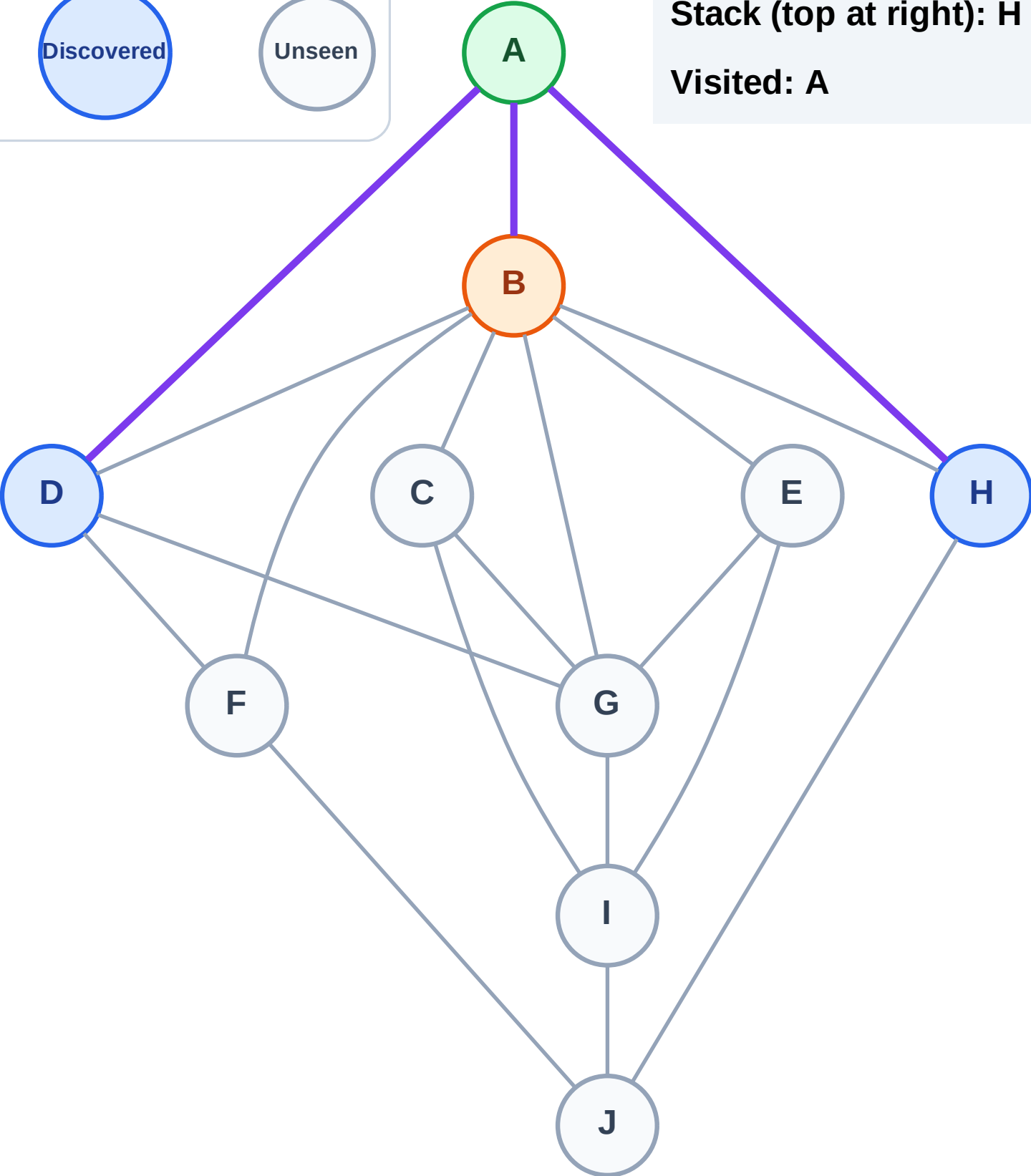
Complete

Processing

Discovered

Unseen

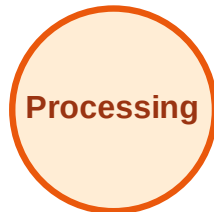
Stack (top at right): H | D
Visited: A



DFS Traversal

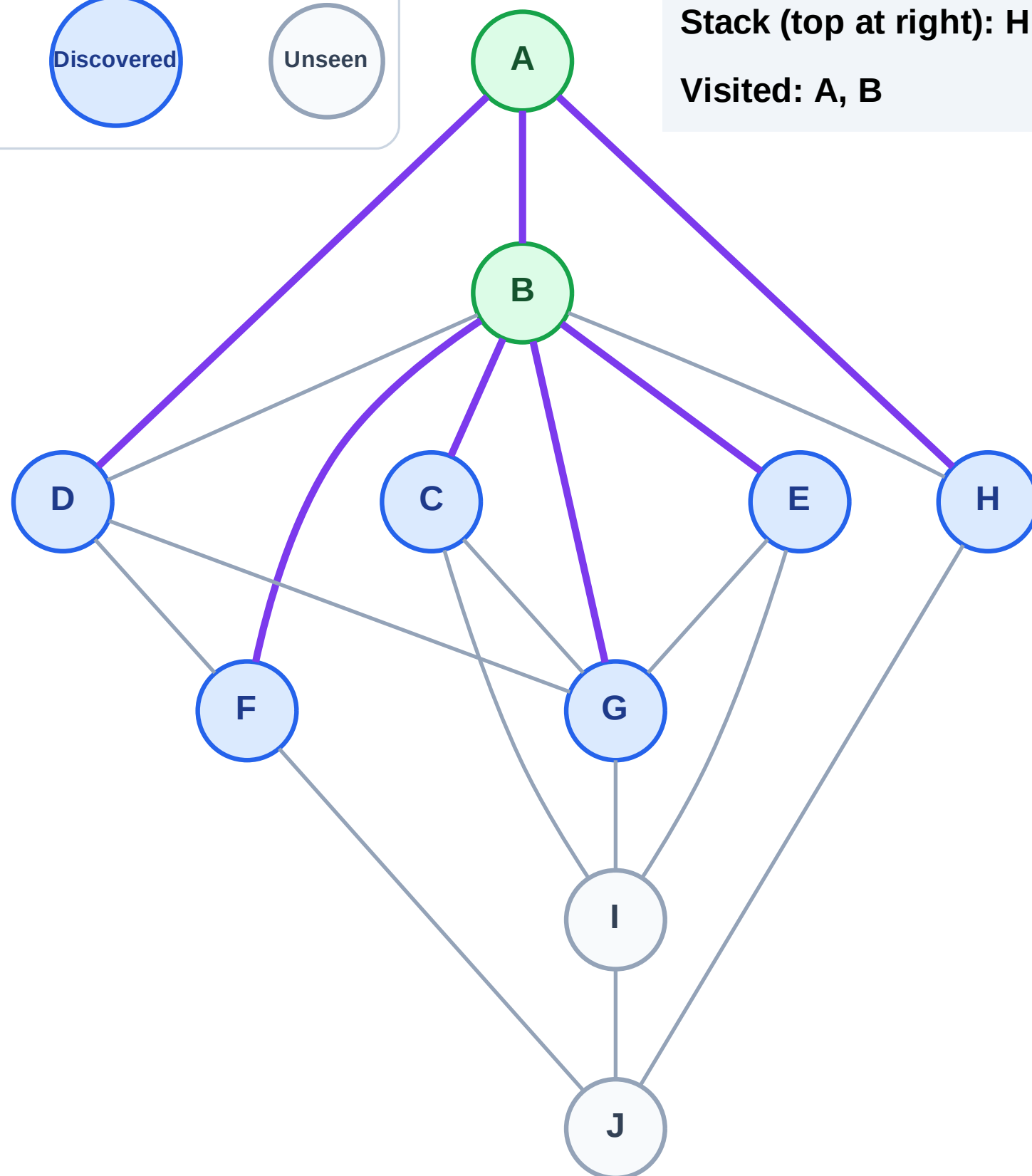
Step 5: Discover G, F, E, C from B

Legend



Stack (top at right): H | D | G | F | E | C

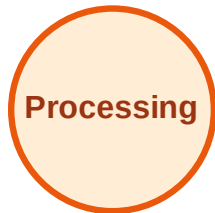
Visited: A, B



DFS Traversal

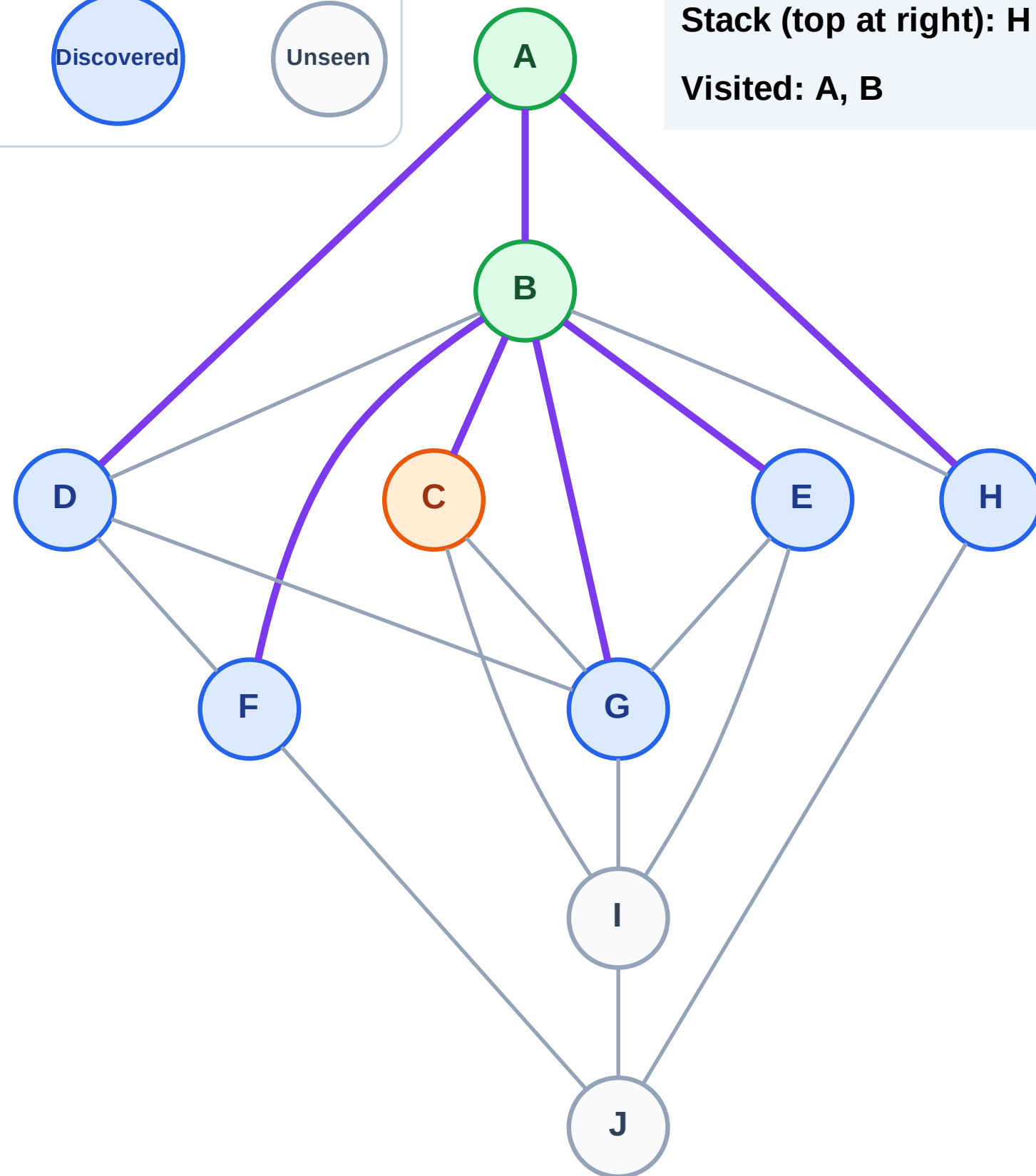
Step 6: Process node C

Legend



Stack (top at right): H | D | G | F | E

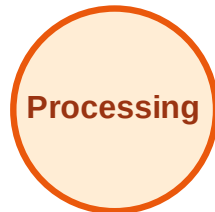
Visited: A, B



DFS Traversal

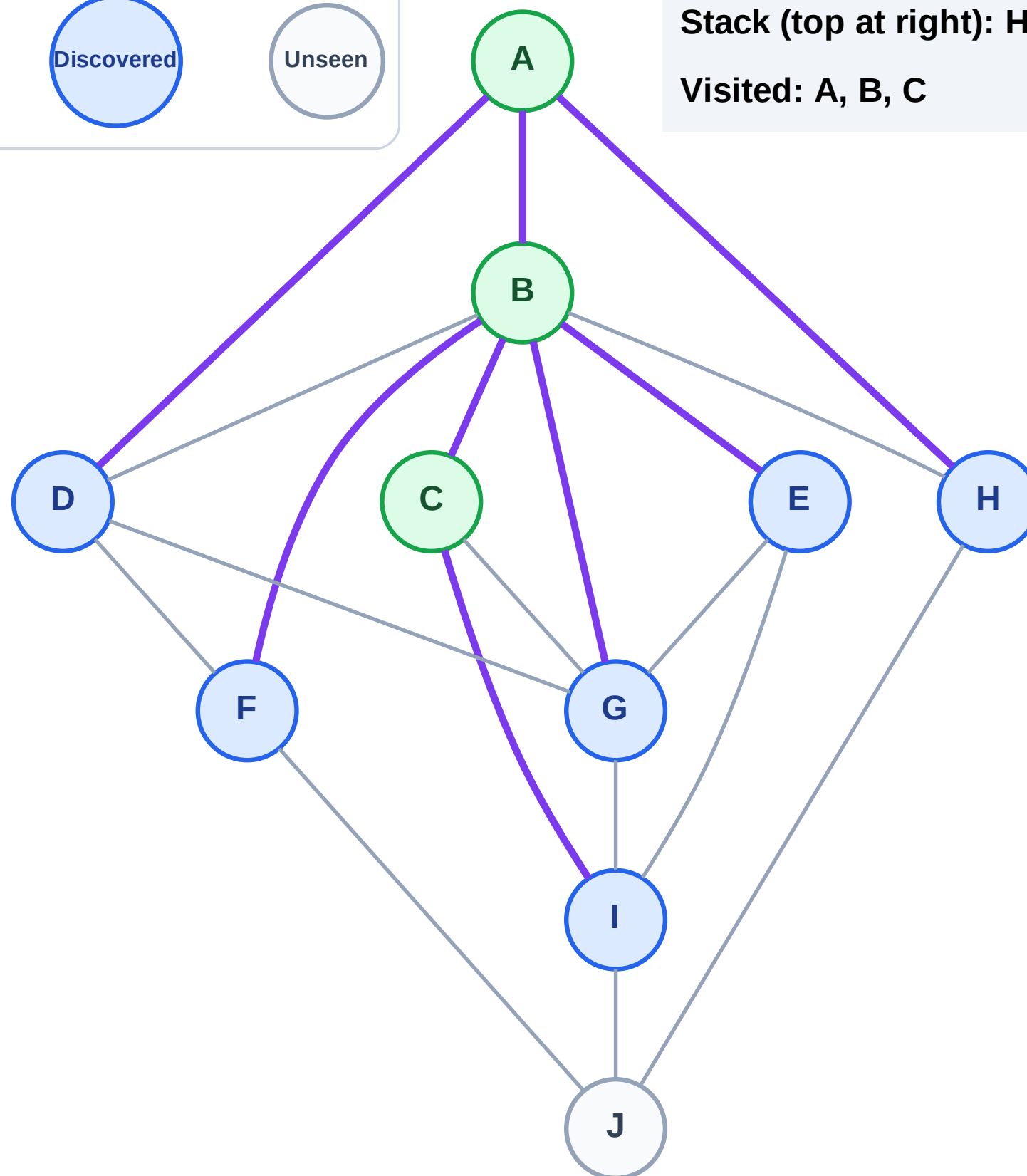
Step 7: Discover I from C

Legend



Stack (top at right): H | D | G | F | E | I

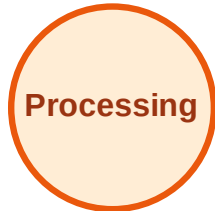
Visited: A, B, C



DFS Traversal

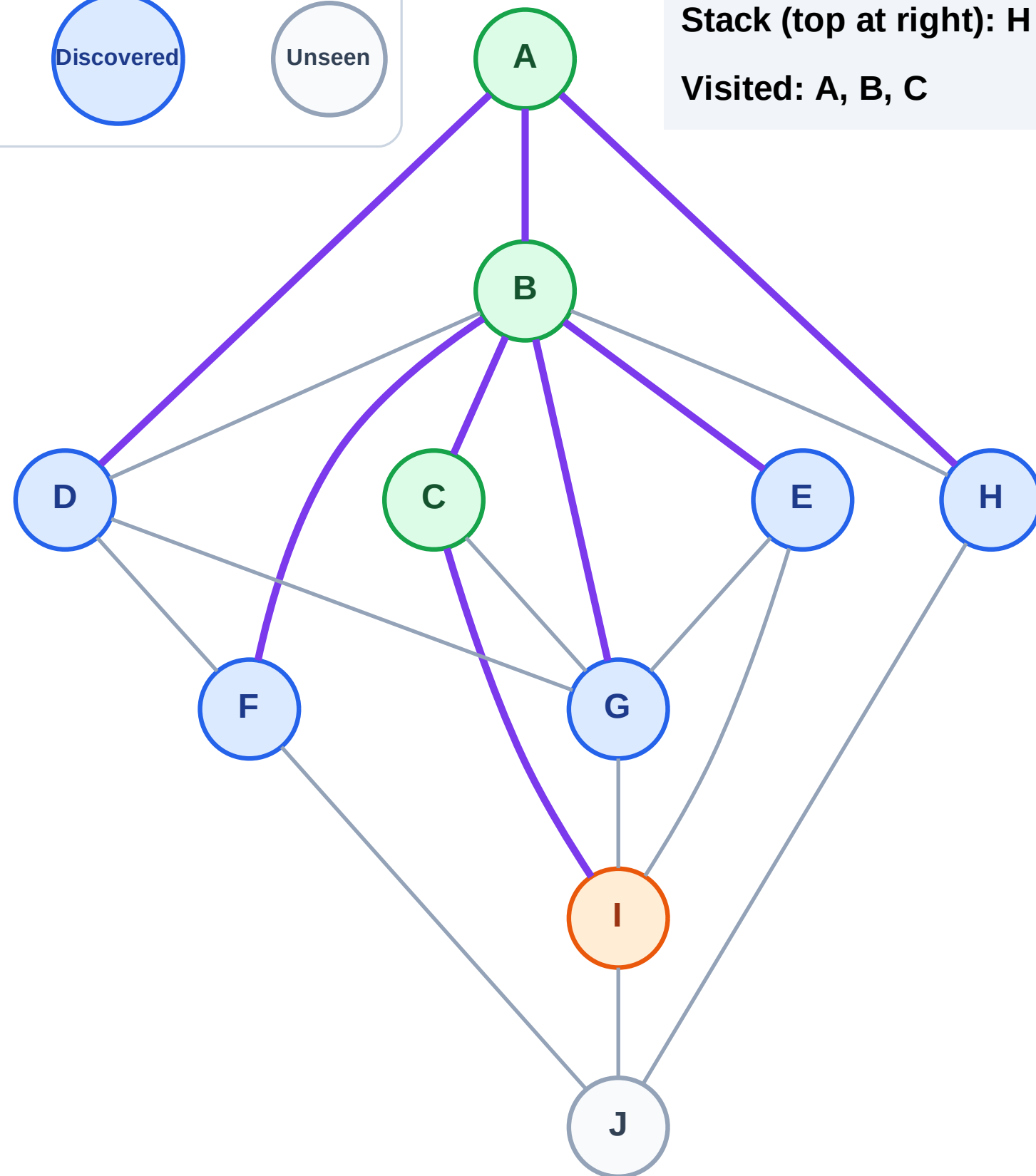
Step 8: Process node I

Legend



Stack (top at right): H | D | G | F | E

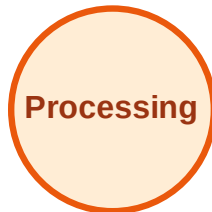
Visited: A, B, C



DFS Traversal

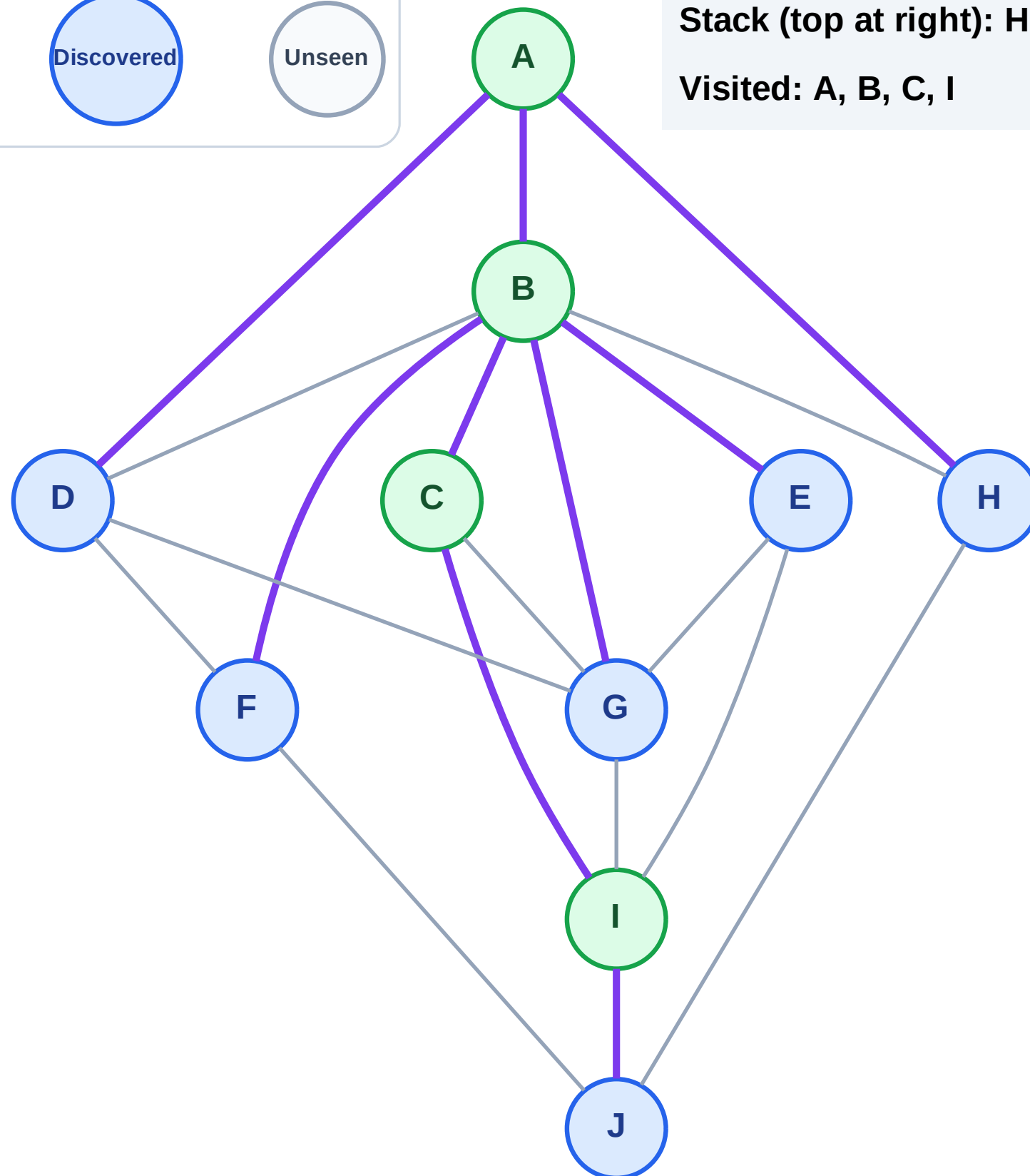
Step 9: Discover J from I

Legend



Stack (top at right): H | D | G | F | E | J

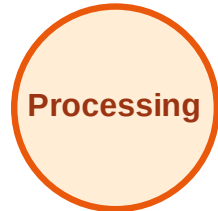
Visited: A, B, C, I



DFS Traversal

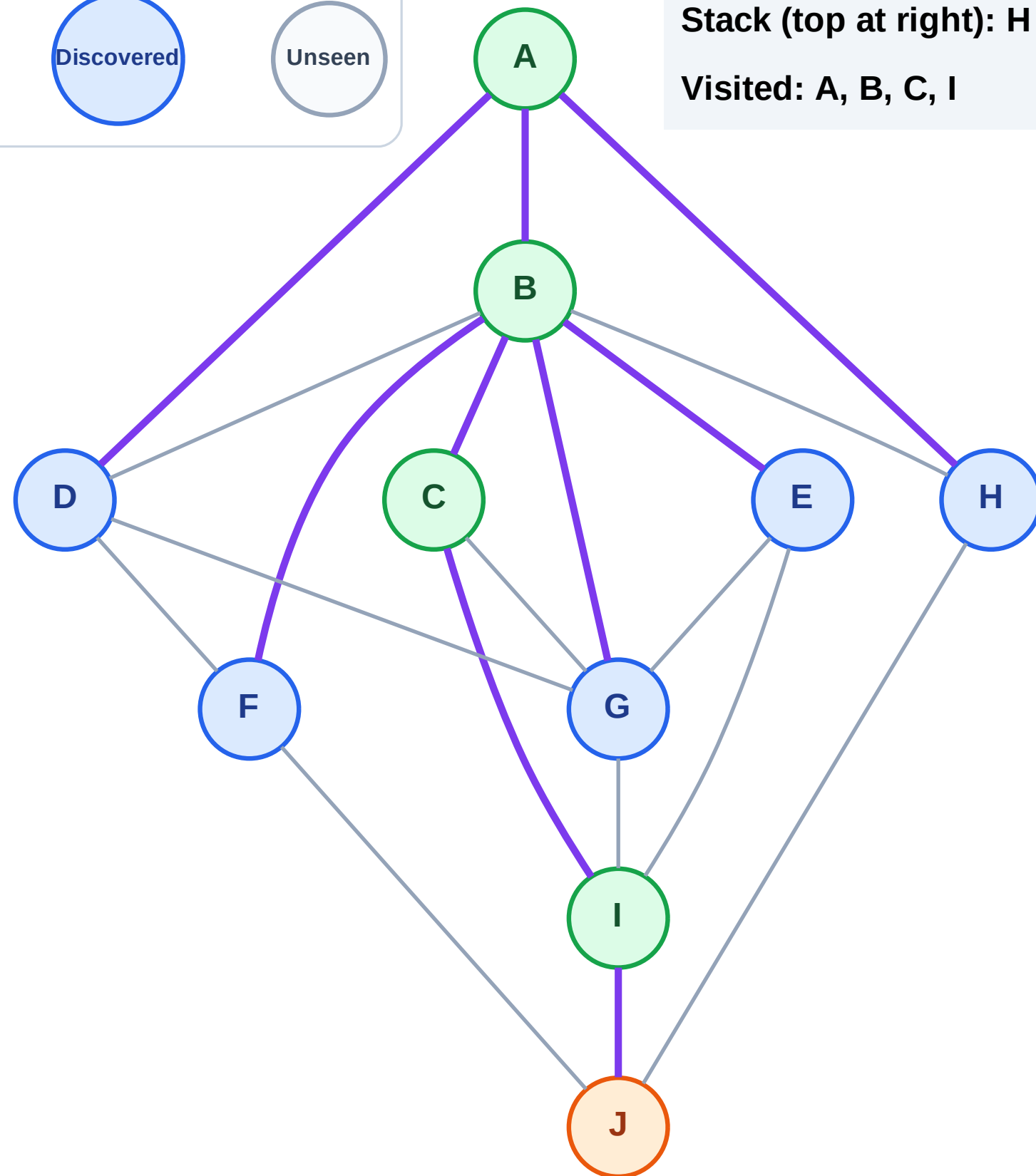
Step 10: Process node J

Legend



Stack (top at right): H | D | G | F | E

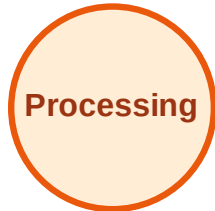
Visited: A, B, C, I



DFS Traversal

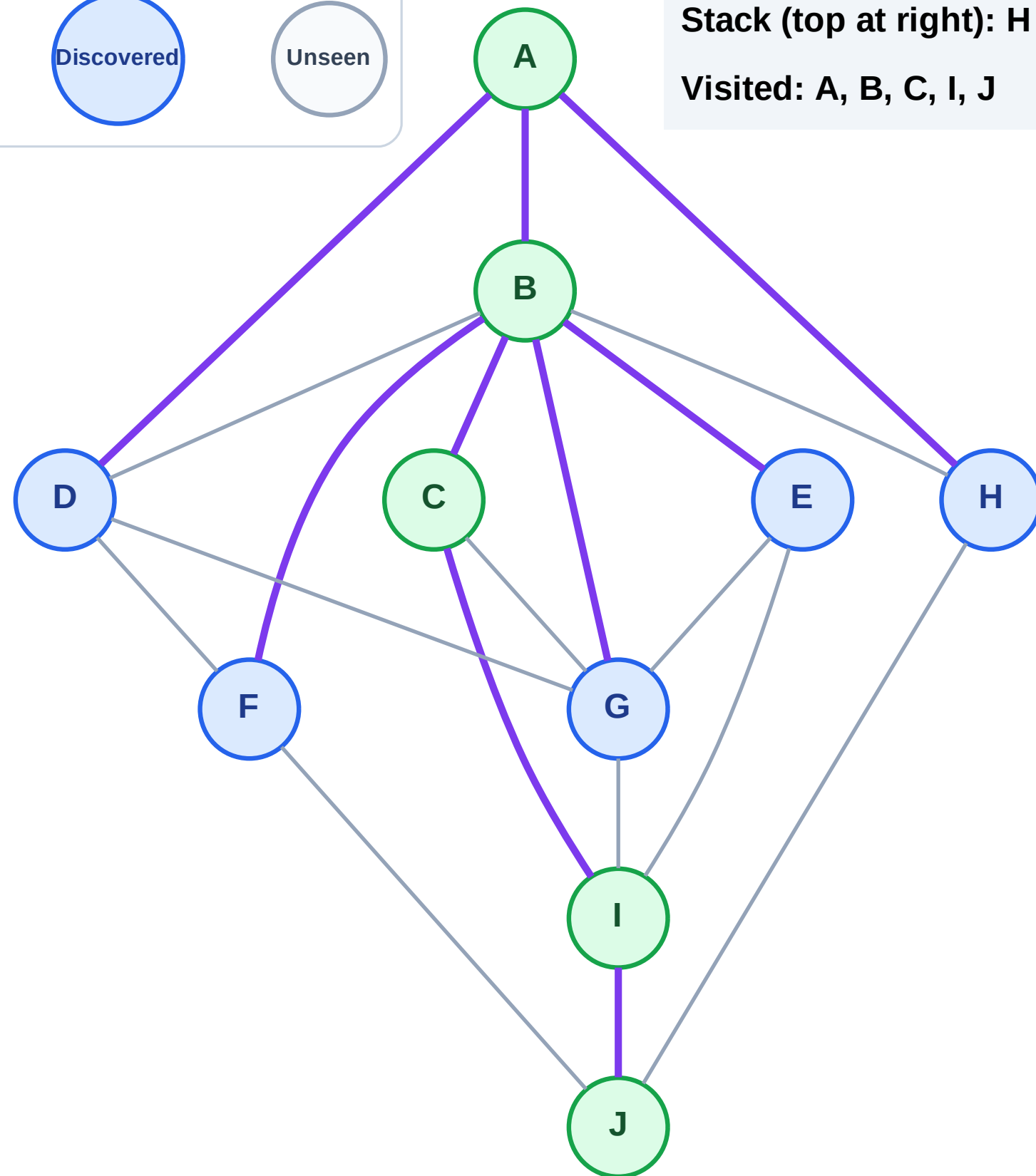
Step 11: No new neighbors from J

Legend



Stack (top at right): H | D | G | F | E

Visited: A, B, C, I, J



DFS Traversal

Step 12: Process node E

Legend

Complete

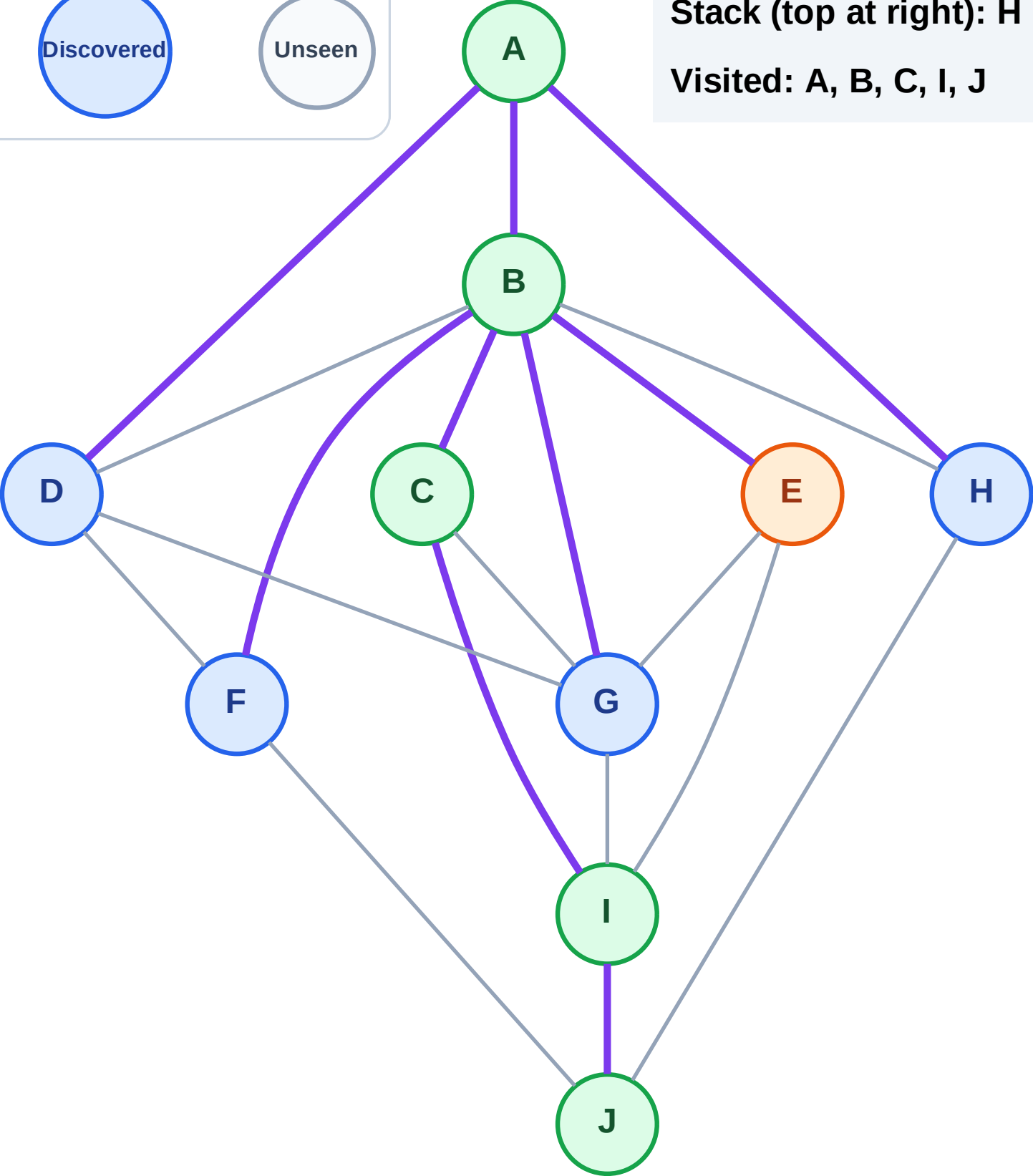
Processing

Discovered

Unseen

Stack (top at right): H | D | G | F

Visited: A, B, C, I, J



DFS Traversal

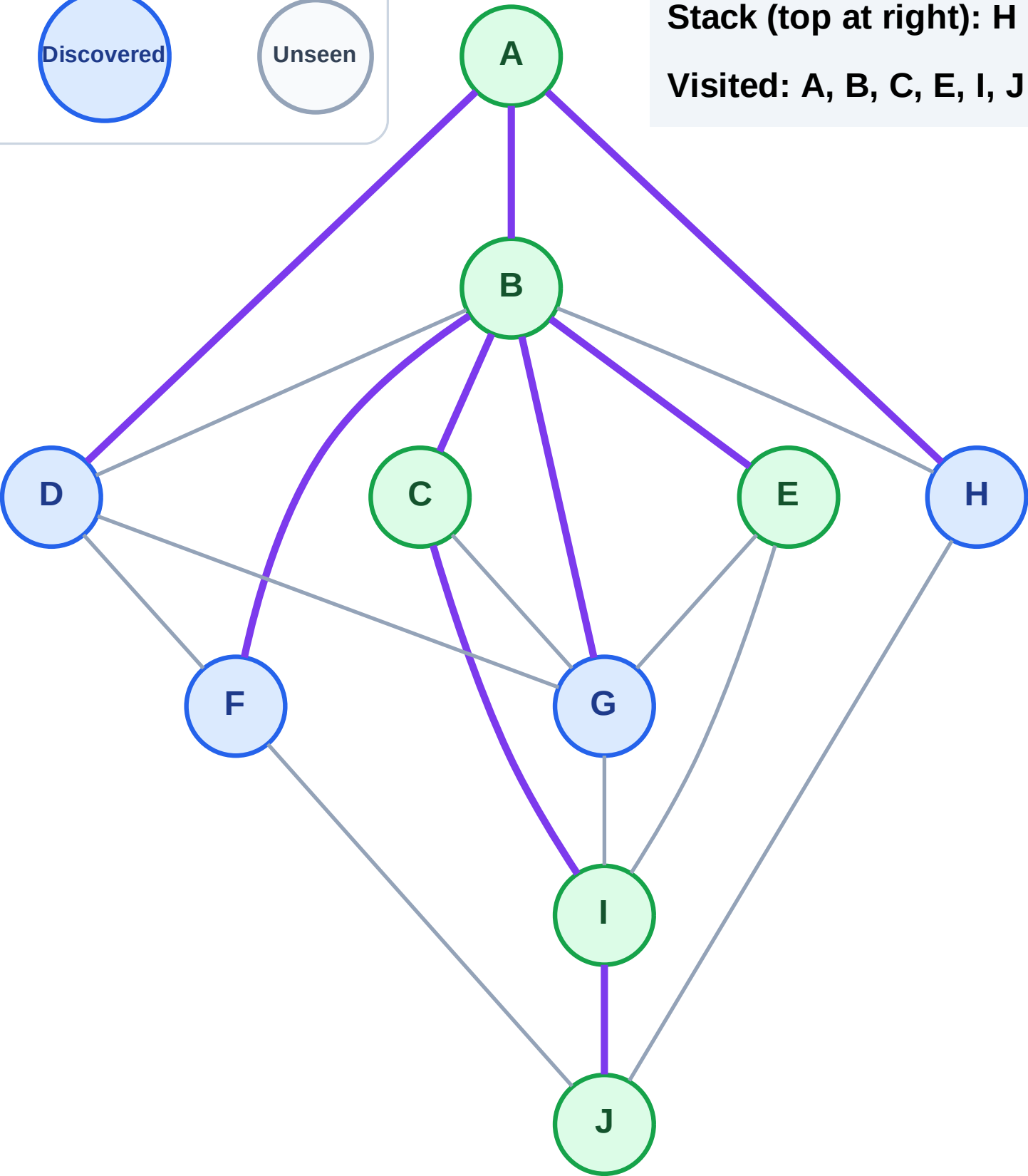
Step 13: No new neighbors from E

Legend



Stack (top at right): H | D | G | F

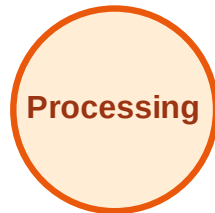
Visited: A, B, C, E, I, J



DFS Traversal

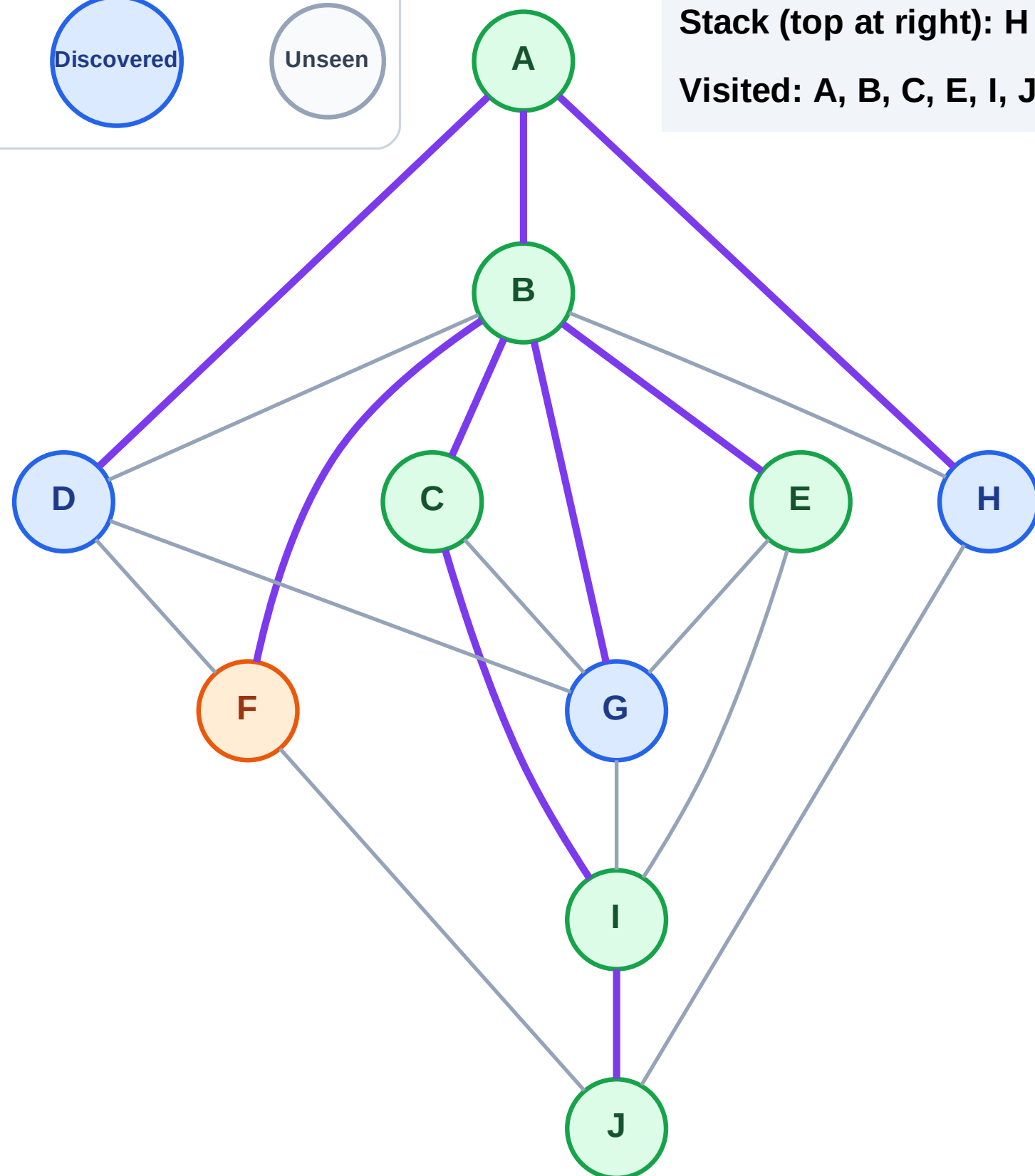
Step 14: Process node F

Legend



Stack (top at right): H | D | G

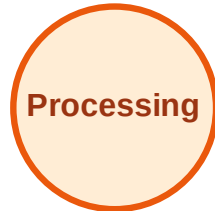
Visited: A, B, C, E, I, J



DFS Traversal

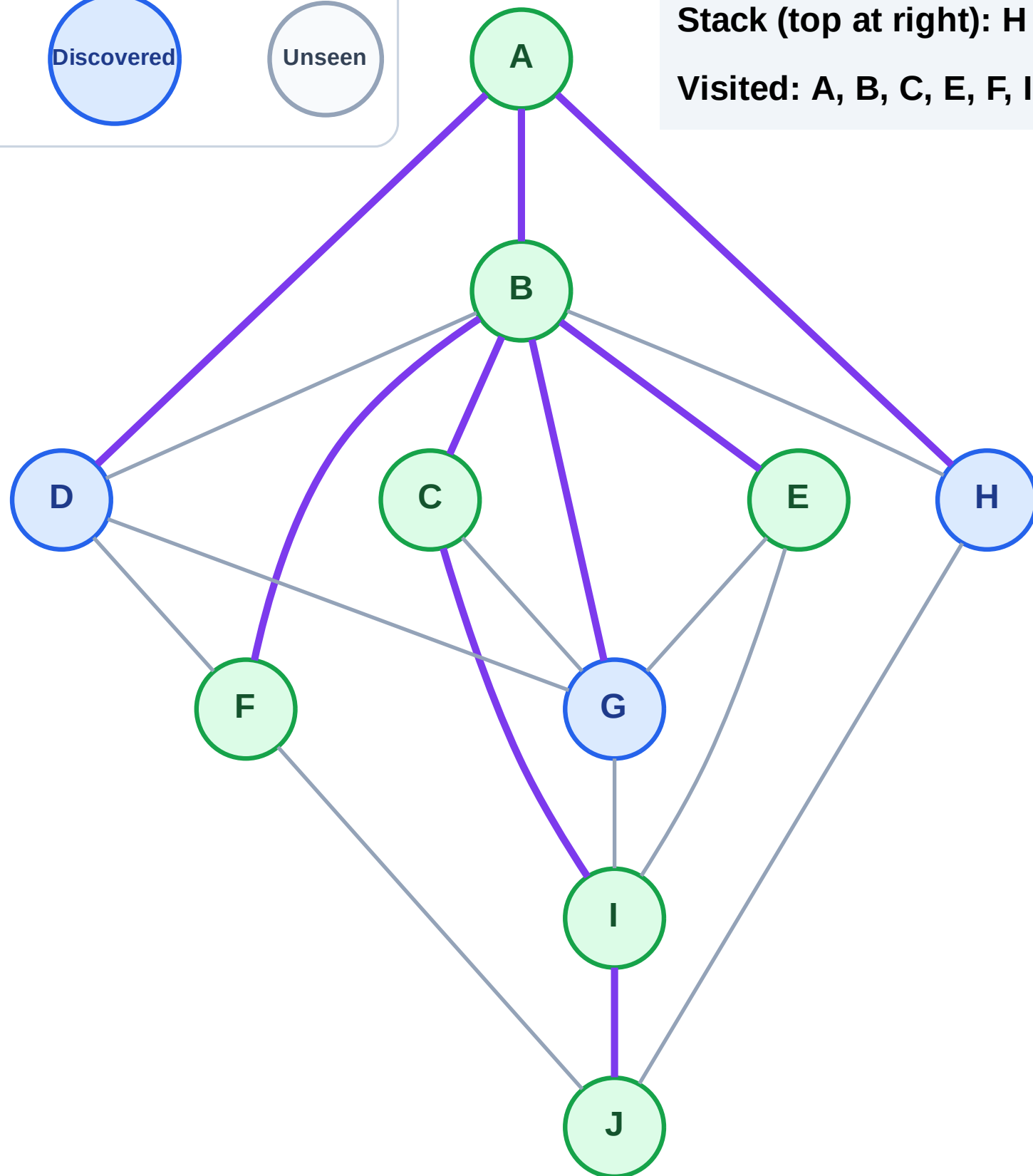
Step 15: No new neighbors from F

Legend



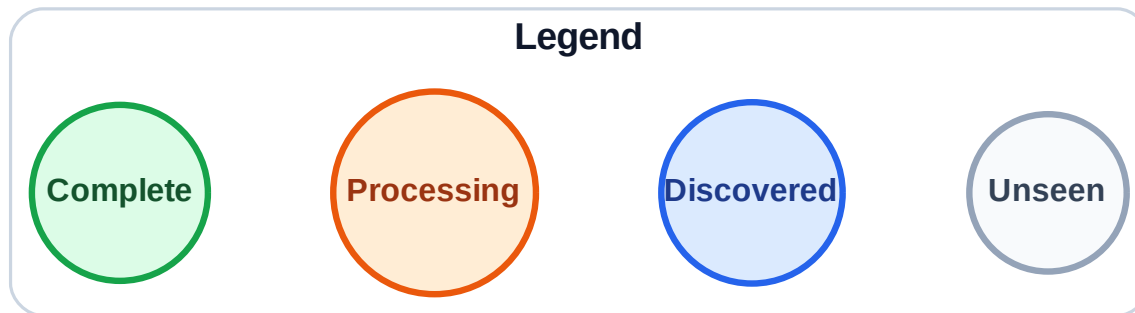
Stack (top at right): H | D | G

Visited: A, B, C, E, F, I, J



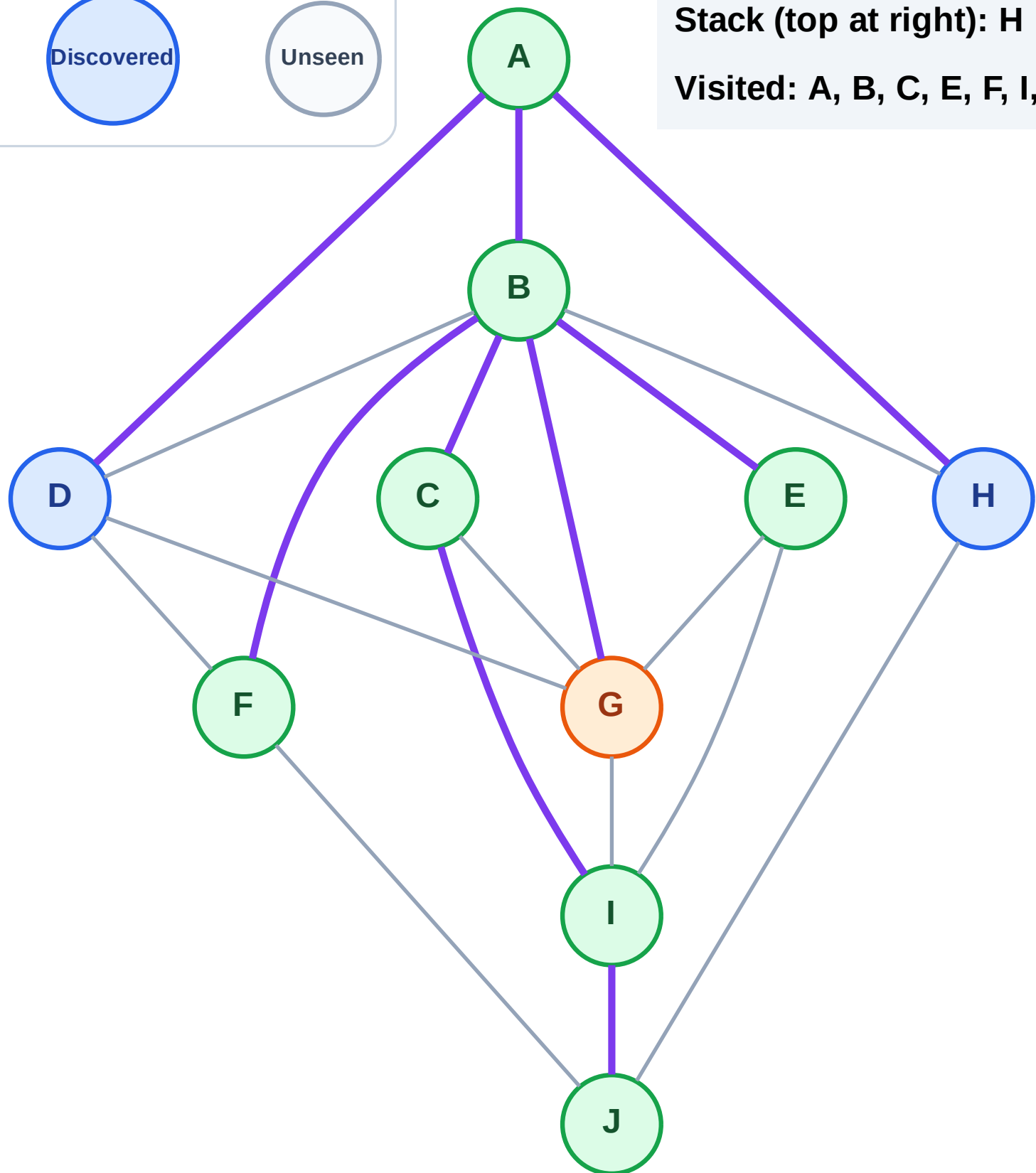
Step 16: Process node G

Step 16: Process node G



Stack (top at right): H | D

Visited: A, B, C, E, F, I, J



DFS Traversal

Step 17: No new neighbors from G

Legend

Complete

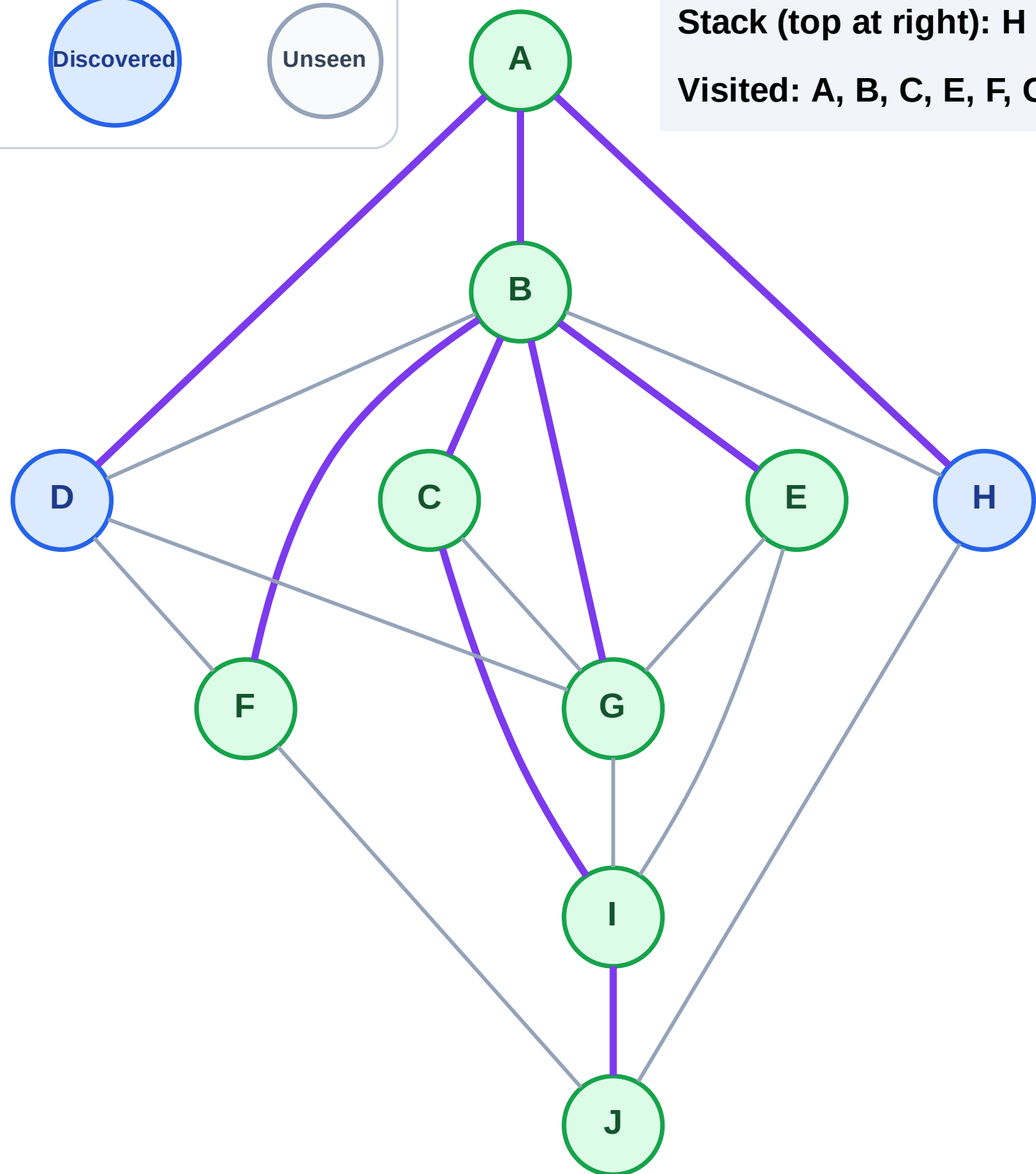
Processing

Discovered

Unseen

Stack (top at right): H | D

Visited: A, B, C, E, F, G, I, J



DFS Traversal

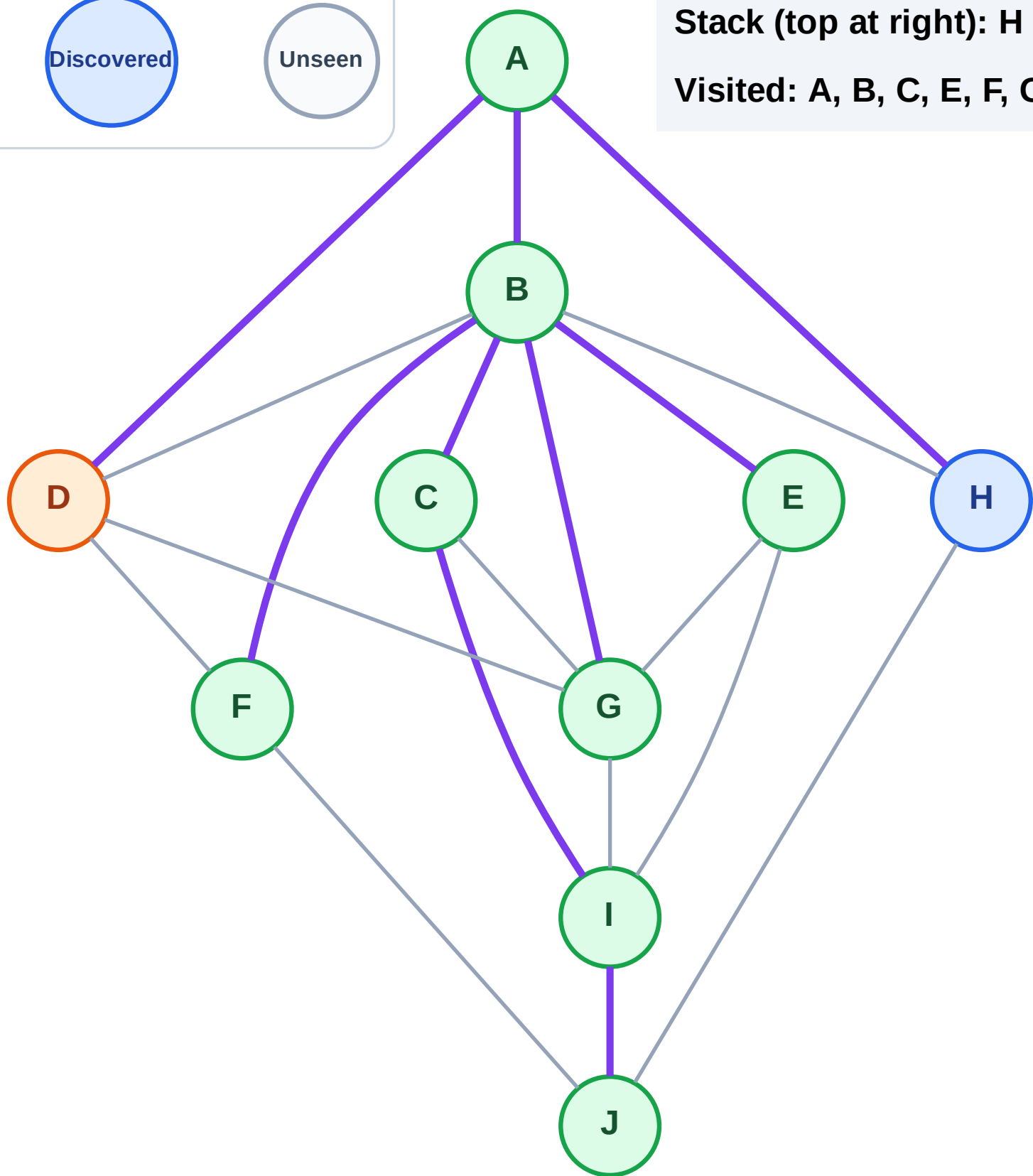
Step 18: Process node D

Legend



Stack (top at right): H

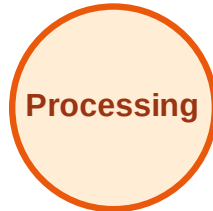
Visited: A, B, C, E, F, G, I, J



DFS Traversal

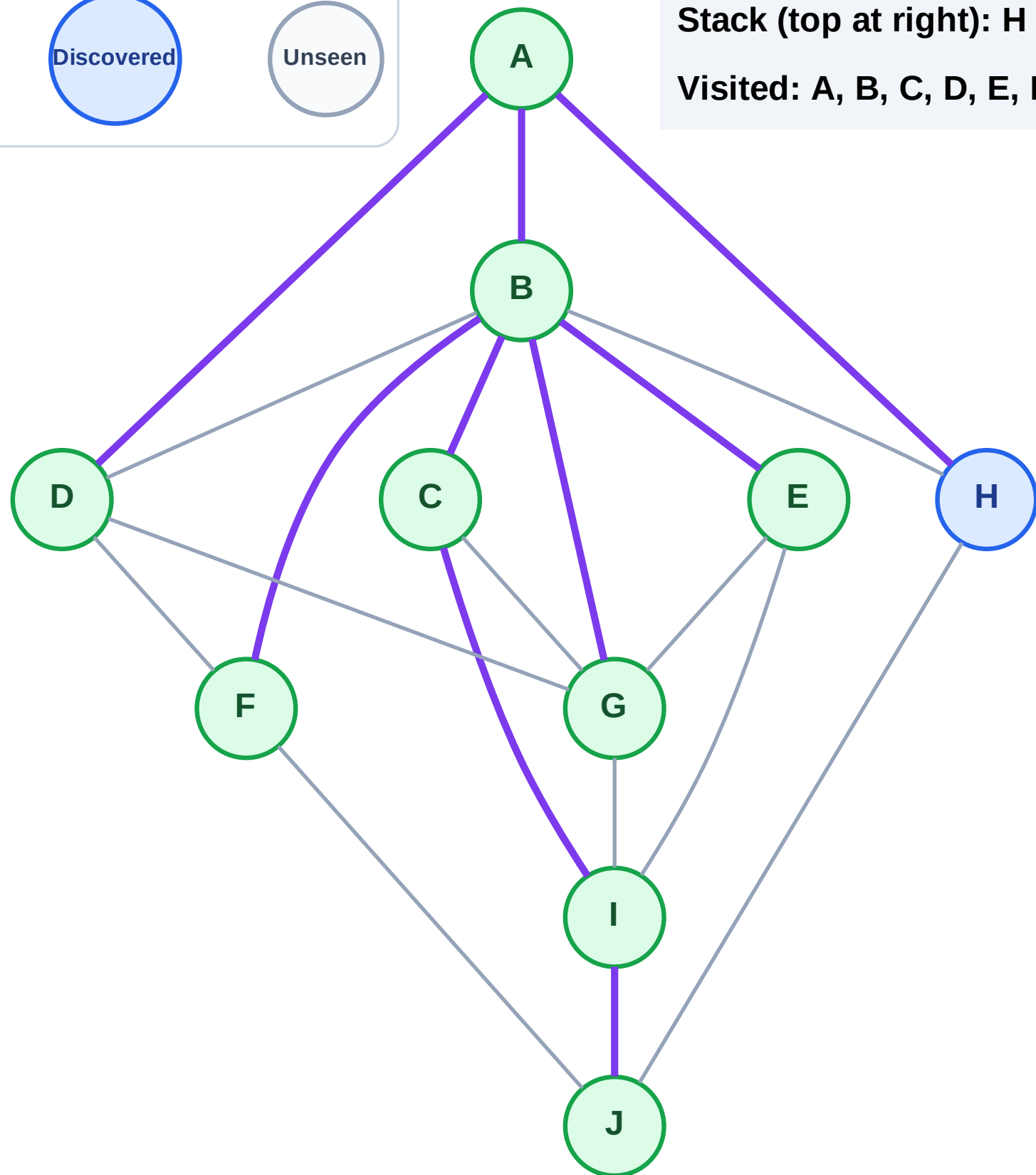
Step 19: No new neighbors from D

Legend



Stack (top at right): H

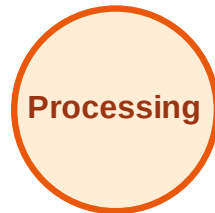
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

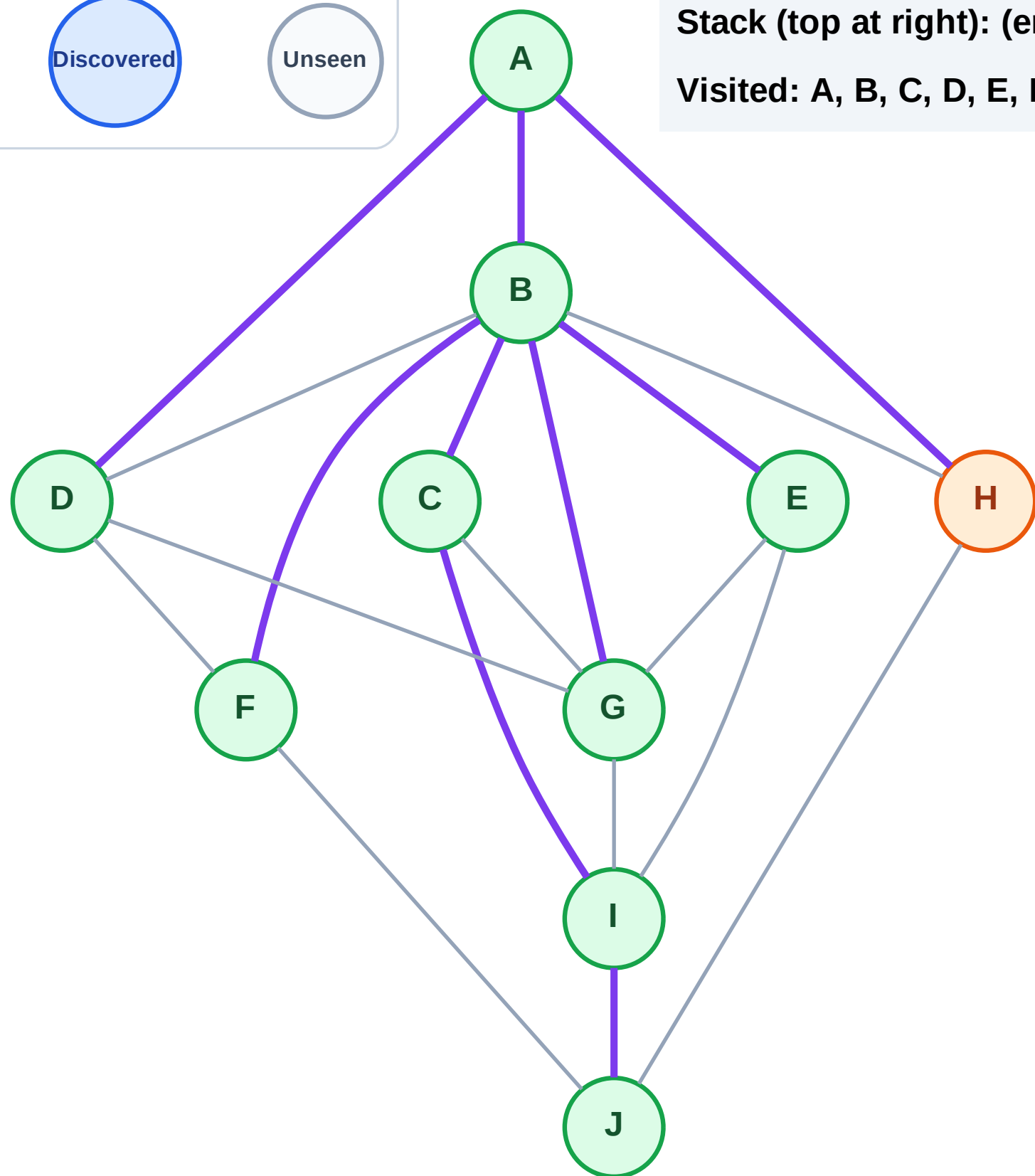
Step 20: Process node H

Legend



Stack (top at right): (empty)

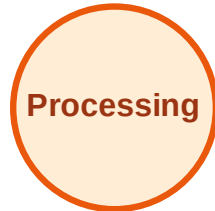
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

